



DR ACADEMY

DO RIGHT FOR GENUINE EDUCATION

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NEET EXAMINATION - 2026

DATE : 21-06-2026

CODE - 60

TIME : 02:00 PM TO 05:15 PM

LONG TERM COACHING

1ST BATCH FOR NEET - 2027

25TH STARTS ON

JUNE - 2026

NEET - 2025 TOPPERS

 607 MARKS AFNAAN PASHA App No. 250410602893	 605 MARKS RAKESH KUMBAR App No. 250410893858	 589 MARKS M LEENA REDDY App No. 250410154513	 588 MARKS JAYANTH K S App No. 250410619327
 581 MARKS CHANDANA REDDY N G App No. 250410232370	 581 MARKS BHAGYASHRI PATTAR App No. 250410153733	 579 MARKS KOTTAPALLI ABDUL KALAM App No. 250410865706	 577 MARKS RAJMANE TRUPTI RAHUL App No. 250410324087
 575 MARKS CHANDRIKA SAJJANAR App No. 250410695592	 572 MARKS VIJAY M App No. 250410629648	 571 MARKS ANMOL SHEGUNASI App No. 250410421533	 569 MARKS AKSHATA GURAPPA J App No. 250411245807
 562 MARKS SAMEERAHMAD SAIFANSAB App No. 250411621485	 562 MARKS RAKSHITH KUMAR S App No. 250410518308	 560 MARKS M L THANMAYI ANNAPOORNA App No. 250411113373	 558 MARKS SHREE GOURI App No. 250410227908
 555 MARKS YOGITHA D App No. 250410071616	 555 MARKS SURAJ K R App No. 250410942156	 555 MARKS NAGASAI CH App No. 250410870699	 553 MARKS PRAJWAL N S App No. 250410415967
			 553 MARKS CHINMAYI SARANGAMATH App No. 250410267618

NEET TOPPERS OF DR ACADEMY

NEET - 2024 JIPMER - PUDUCHERRY  706 MARKS VENKATA SURYA TEJA GUDURI	NEET - 2021 JIPMER - PUDUCHERRY  695 MARKS LAKSHMAN REDDY B V	NEET - 2024 BMC - BANGALORE  690 MARKS ABHISHEK A BADAGOUDAR	NEET - 2024 BMC - BANGALORE  686 MARKS RAHUL KADLAGOND	NEET - 2024 BIMC - BELAGAVI  686 MARKS AMEERMUSADDIQ SANADI	NEET - 2024 KIMS - HUBLI  686 MARKS ADITYA SAKRI
NEET - 2024 BMC - BANGALORE  681 MARKS UTTAM SUBHASH HUKKERI	NEET - 2024 SABVIMS - BENGALURU  681 MARKS MOHAMMED NAASIRUDEEN D	NEET - 2024 MMCRI - MYSORE  680 MARKS SUHAS L KORABU	NEET - 2023 BMC, BANGALORE  680 MARKS SHRAVAN REDDY C N	NEET - 2023 JIPMER PUDUCHERRY  677 MARKS KAMALIKA CHALLA	NEET - 2022 BIMC, BELAGAVI  677 MARKS SACHI KALLOLI
NEET - 2023 AIIMS, NAGPUR  676 MARKS MOHAMMED SULEMAN	NEET - 2022 AIIMS, BHOPAL  675 MARKS CHATHUSH GOWDA D S	NEET - 2022 BMC, BANGALORE  672 MARKS CHANDANA D	NEET - 2022 BMC, BANGALORE  672 MARKS SOWRAV B	NEET - 2024 ESIMC - BANGALORE  671 MARKS ZOYA FIRDOUSE	NEET - 2024 MMCRI - MYSORE  671 MARKS GIRISH J PARAMAGOND
NEET - 2024 KIMS - HUBLI  670 MARKS SNEHA SUBHAS PATIL	NEET - 2024 KIMS - HUBLI  666 MARKS RAJIV BHEEMASHANKAR CHOUDHARI	NEET - 2024 KIMS - HUBLI  665 MARKS C SATHYAM	NEET - 2021 BMC, BANGALORE  665 MARKS RASHMI PATIL	NEET - 2020 BMC, BANGALORE  665 MARKS JAYANTH L S	NEET - 2024 GIMS - GULBARGA  663 MARKS MOHAMMED ZEESHAN
NEET - 2024 MIMS - MANDYA  662 MARKS POOJA U	NEET - 2024 MMCRI - MYSORE  662 MARKS SAMPAT GOPAL GOKAK	NEET - 2022 BMC, BANGALORE  662 MARKS SIDDHARTH A S	NEET - 2024 KIMS - HUBLI  661 MARKS DIVYA M YALIGAR	NEET - 2023 JIPMER PUDUCHERRY  661 MARKS SHASHANK SURAPOOR	NEET - 2024 BIMC - BELAGAVI  660 MARKS ANMOL R KOTNAL
NEET - 2024 KIMS - BANGALORE  660 MARKS ADHITHYA SUDARSAN GOKHALE	NEET - 2024 MMCRI - MYSORE  660 MARKS SHRIDHAR BIRADAR	NEET - 2024 SABVIMS - BENGALURU  660 MARKS POOJA N	NEET - 2024 KIMS - HUBLI  660 MARKS TARUN N	NEET - 2021 KIMS, HUBLI  660 MARKS NACHIKET KEMPANNA	NEET - 2024 KIMS - HUBLI  657 MARKS SINDHU VADAVADI
NEET - 2024 MIMS - MANDYA  657 MARKS S SUHAS	NEET - 2024 SABVIMS - BENGALURU  657 MARKS SHREEHARINATH A B	NEET - 2024 GMC - SAMBAHAJINAGAR  656 MARKS PRATEEK SUBHASH TOPINATTI	NEET - 2024 KIMS - HUBLI  656 MARKS YASEEN MULLA	NEET - 2024 MIMS - MANDYA  656 MARKS KUSUMA M	NEET - 2024 SIMS - SHIMOGA  656 MARKS NAGAVARDHAN MR
NEET - 2024 VIMS - BELLARY  656 MARKS N POOJITHA SHREE	NEET - 2023 BMC, BANGALORE  656 MARKS BHANU PRAKASH D M	NEET - 2021 AIIMS, HYDERABAD  656 MARKS VARUN KAJAGAR	NEET - 2024 AIIMS, HYDERABAD  655 MARKS AMODH NAIK	NEET - 2024 BIMC - BELAGAVI  655 MARKS SAKSHI CHANDRASHEKHAR YALARADDI	NEET - 2024 HIMS - HASSAN  655 MARKS NITYA REDDY C
NEET - 2024 SIMS - SHIMOGA  655 MARKS MAHIPAL SINGH	NEET - 2024 VIMS - BELLARY  654 MARKS NIVEDITA	NEET - 2023 BMC, BANGALORE  654 MARKS ABHISHEK V G	NEET - 2021 BMC, BANGALORE  654 MARKS KUMARESH HIREMATH	NEET - 2024 BIMC - BELAGAVI  653 MARKS SIDDANAGOUDA S PATIL	NEET - 2024 KIMS - HUBLI  653 MARKS VINDHYA B G
NEET - 2024 BIMC - BELAGAVI  652 MARKS BHOOMIKA BAYAKOL	NEET - 2024 VIMS - BELLARY  652 MARKS CHIRANTHAN J	NEET - 2024 VIMS - BELLARY  652 MARKS B K V KARTHIKEYA	NEET - 2024 KIMS - HUBLI  651 MARKS SHASHANK VALIMARAD	NEET - 2024 KIMS - HUBLI  650 MARKS PARASHURAM KYADIGGERI	NEET - 2024 MIMS - MANDYA  650 MARKS PREETHAM K M

DAY & RESIDENTIAL
SEPARATE HOSTEL FOR BOYS & GIRLS

LTM ADMISSIONS OPEN

PHYSICS QUESTION PAPER ANALYSIS

CHAPTER

NO. OF QUESTIONS

I PUC

UNITS AND MEASUREMENT

2

MOTION IN A STRAIGHT LINE

1

MOTION IN A PLANE

0

LAWS OF MOTION

0

WORK, ENERGY & POWER

3

SYSTEM OF PARTICLES & ROTATIONAL MOTION

3

GRAVITATION

2

MECHANICAL PROPERTIES OF SOLIDS

0

MECHANICAL PROPERTIES OF FLUIDS

2

THERMAL PROPERTIES OF MATTER

1

THERMODYNAMICS

3

KINETIC THEORY

1

OSCILLATIONS

2

WAVES

1

II PUC

ELECTRIC CHARGES AND FIELDS

1

ELECTROSTATIC POTENTIAL AND CAPACITANCE

3

CURRENT ELECTRICITY

1

MOVING CHARGES AND MAGNETISM

2

MAGNETISM AND MATTER

0

ELECTROMAGNETIC INDUCTION

3

ALTERNATING CURRENT

1

ELECTROMAGNETIC WAVES

2

RAY OPTICS AND OPTICAL INSTRUMENTS

3

WAVE OPTICS

0

DUAL NATURE OF RADIATION AND MATTER

3

ATOMS

1

NUCLEI

2

SEMI CONDUCTOR ELECTRONICS

2

PHYSICS

1. An ac voltage $V = 220 \sin(2 \times 10^2 t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :

(Given : $L = 10 \text{ mH}$, $C = 25 \mu\text{F}$, $R = 100 \Omega$)

- 1) 11.0 A 2) 22.0 A 3) 2.2 A 4) 5.5 A

Ans. 3

Sol. $X_L = X_C$

$$I_0 = \frac{V_0}{R} = \frac{220}{100} = 2.2 \text{ A}$$

2. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B , respectively, then the correct option is:

- 1) $n_A = \frac{1}{4} n_B$ 2) $n_A = \frac{1}{2} n_B$
3) $n_A = n_B$ 4) $n_A = 2n_B$

Ans. 2

Sol. $\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$

$$n \propto \frac{1}{\lambda d^2}$$

$$\frac{n_A}{n_B} = \frac{\lambda_B \left(\frac{D_B}{D_A} \right)^2}{\lambda_A \left(\frac{D_B}{D_A} \right)^2} = 2 \left(\frac{1}{2} \right)^2 = \frac{1}{2}$$

$$n_A = \frac{1}{2} n_B$$

3. A cylindrical cork of uniform density floats in a liquid of density ρ_1 . If the cork is depressed slightly and released, it oscillates harmonically with time period T . If the same cork floats in another liquid of density ρ_2 , then the similar oscillation has time period $2T$. The value of ρ_2 / ρ_1 is

- 1) $1/2$ 2) $1/4$ 3) 4 4) 2

Ans. 2

Sol. $T \propto \frac{1}{\sqrt{\rho}}$

$$T^2 \propto \frac{1}{\rho}$$

$$\frac{\rho_2}{\rho_1} = \left(\frac{T_1}{T_2} \right)^2 = \left(\frac{T}{2T} \right)^2 = \frac{1}{4}$$

4. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is $k \text{ Nm}^{-1}$. At a given instant, the extension of the spring is x meter and the speed of the particle is $v \text{ ms}^{-1}$. On the x - v plane, if the graph of v as a function of x is a circle, then the correct option is

- 1) $k = m^2$ 2) $k = \sqrt{m}$ 3) $k = \frac{1}{m}$ 4) $k = m$

Ans. 4

Sol. $E = \frac{1}{2} mv^2 + \frac{1}{2} kx^2$

$$\frac{\frac{1}{2} mv^2}{E} + \frac{\frac{1}{2} kx^2}{E} = 1$$

$$\frac{v^2}{2E/m} + \frac{x^2}{2E/k} = 1$$

Given it's circle

$$\frac{2E}{m} = \frac{2E}{k}$$

$$m = k$$

5. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60K to 50K. The work done by the gas in the process is :

(Take the universal gas constant as $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

- 1) 124.5 J 2) 166 J 3) 41.5 J 4) 83 J

Ans. 1

Sol. $\gamma = 5/3$

$$60 \text{ K} \rightarrow 50 \text{ K}$$

$$W = \frac{nR(T_1 - T_2)}{\gamma - 1} = 124.5 \text{ J}$$

6. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum	Applications
P. Microwave	I. For purifying the water
Q. UV rays	II. For warming the food
R. Gamma rays	III. For AM and FM communication systems
S. Radio wave	IV. For treating the Cancer cells

The **correct** option is :

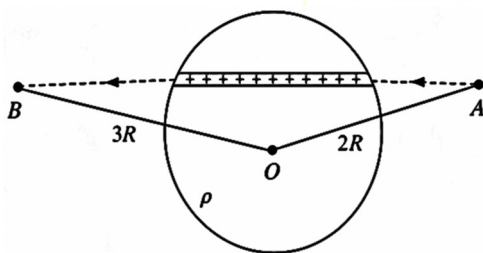
- 1) P-II, Q-I, R-IV, S-III
- 2) P-II, Q-IV, R-III, S-I
- 3) P-I, Q-II, R-III, S-IV
- 4) P-I, Q-IV, R-II, S-III

Ans. 1

Sol. P-II, Q-I, R-IV, S-III

7. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R , having uniform positive charge density ρ , as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is:

(ϵ_0 is the permittivity of vacuum)



- 1) 9
- 2) 18
- 3) 2
- 4) 6

Ans. 2

Sol.
$$V = \frac{1}{4\pi\epsilon_0} \frac{\rho \left(\frac{4}{3} \pi R^3 \right)}{r} = \frac{\rho R^3}{3\epsilon_0 r}$$

$$V_1 = \frac{\rho R^3}{3\epsilon_0 (2R)} = \frac{\rho R^2}{6\epsilon_0}$$

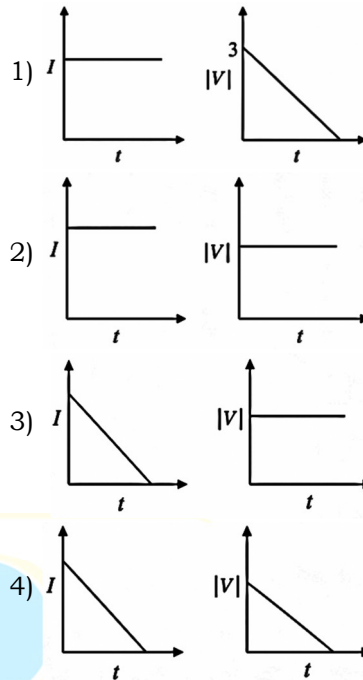
$$V_2 = \frac{\rho R^3}{3\epsilon_0 (3R)} = \frac{\rho R^2}{9\epsilon_0}$$

$$W = \frac{\rho R^2}{\epsilon_0} \left(\frac{1}{6} - \frac{1}{9} \right)$$

$$= \frac{\rho R^2}{\epsilon_0} \left(\frac{3}{54} \right) = \frac{\rho R^2}{18\epsilon_0}$$

$$\therefore n = 18$$

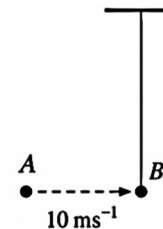
8. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t , then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :



Ans. 3

Sol. When power decreases current decreases with time, but stopping potential remain same.

9. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed 10 ms^{-1} hits bob B elastically. The bob rises h meter after the collision. Taking the acceleration due to gravity $g = 10 \text{ ms}^{-2}$ and neglecting the size of the bob, the value of h is



- 1) 5
- 2) 2.5
- 3) 8
- 4) 7

Ans. 1

Sol. $m_1 = m_2 = m$

$$u_1 = 10 \text{ m/s}; \quad u_2 = 0 \text{ m/s}$$

When equal masses collided they will exchange their velocities.

$$\therefore v_2 = 10 \text{ m/s}$$

$$\therefore h = \frac{v_2^2}{2g} = \frac{10 \times 10}{2 \times 10} = 5 \text{ m}$$

10. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_p / C_v of the gas is $8/7$, then the value of f is
1) 2 2) 1 3) 4 4) 3

Ans. 3

Sol. Translational $\rightarrow 3$

Rotational $\rightarrow 3$

Vibrational $\rightarrow 2f$

$$\frac{C_p}{C_v} = \gamma = \frac{f_D + 2}{f_D} = \frac{8}{7}$$

$$1 + \frac{2}{f_D} = \frac{8}{7}$$

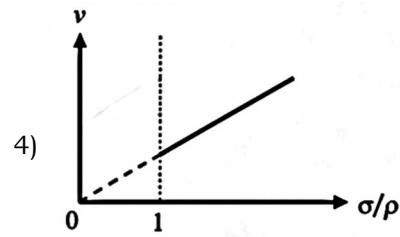
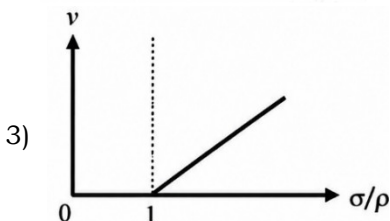
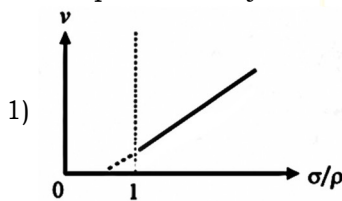
$$\frac{2}{f_D} = \frac{1}{7}$$

$$f_D = 14$$

$$\rightarrow 2f = 14 - 6 = 8$$

$$f = 4$$

11. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :



Ans. 3

$$\text{Sol. } V_T = \frac{2}{9} \frac{r^2 g}{\eta} \sigma \left(\frac{\sigma}{\rho} - 1 \right)$$

12. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to :

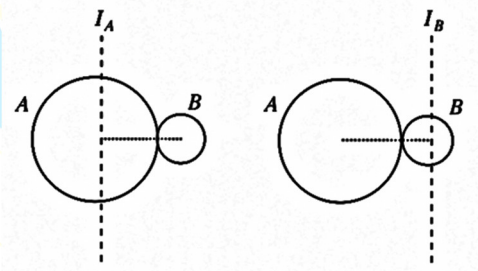
- 1) R^2 2) R^3 3) $R^{1/2}$ 4) $R^{3/2}$

Ans. 4

$$\text{Sol. } T^2 \propto R^3$$

$$T \propto R^{3/2}$$

13. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B . The difference $I_A - I_B$ is



- 1) $(m - M)(R - r)^2$ 2) 0
3) $(M - m)(R + r)^2$ 4) $(m - M)(R + r)^2$

Ans. 4

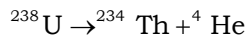
$$\text{Sol. } I_A = \frac{2}{5} MR^2 + \frac{2}{5} mr^2 + m(R + r)^2$$

$$I_B = \frac{2}{5} mr^2 + \frac{2}{5} MR^2 + M(R + r)^2$$

$$I_A - I_B = m(R + r)^2 - M(R + r)^2$$

$$I_A - I_B = (m - M)(R + r)^2$$

14. Consider the following nuclear reaction:



Take masses of ${}^{238}\text{U}$, ${}^{234}\text{Th}$ and ${}^4\text{He}$ as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is :

[Given : 1 u = 931.5 MeV c^{-2}]

- 1) 3736 2) 3740 3) 3726 4) 3730

Ans. 3

Sol. $Q = \Delta m \times 931.5 \text{ MeV}$

$$Q = [M_U - (M_{Th} + M_{He})] \times 931.5$$

$$= 0.004 \times 931.5 \text{ MeV}$$

$$= 3.726 \text{ in MeV} = 3726 \text{ in KeV}$$

15. Consider a fixed uniformly charged insulating sphere with radius R and total charge +Q. A point charge -q ($q \ll Q$) with mass m is released from rest at a distance of 3R from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :

(ϵ_0 is the permittivity of vacuum, neglect gravitational forces)

- 1) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$ 2) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$
3) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$ 4) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

Ans. 1

Sol. From conservation of energy,

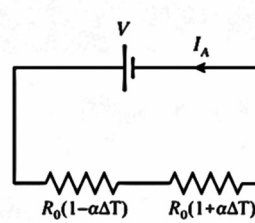
$$U_i + K_i = U_f + K_f$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \left(\frac{-Qq}{3R} \right) + 0 = \frac{1}{4\pi\epsilon_0} \left(\frac{-Qq}{R} \right) + \frac{1}{2}mv^2$$

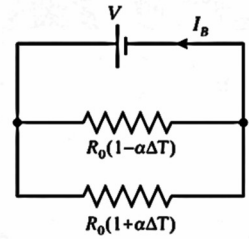
$$\Rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{Qq}{R} \left(\frac{2}{3} \right) = \frac{1}{2}mv^2$$

$$\therefore v = \sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$$

16. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B . At initial temperature, the resistance of the two resistors is R_0 . As the temperature is increased, the correct option that describes the variation of current in these circuits is :



(A)



(B)

- 1) I_A increases while I_B decreases
2) both I_A and I_B remain constant
3) I_A remains constant while I_B increases
4) I_A decreases while I_B increases

Ans. 3

Sol. $R_1 = R_0(1 + \alpha\Delta T)$

$$R_2 = R_0(1 - \alpha\Delta T)$$

Series:

$$R_{eqA} = R_1 + R_2$$

$$= R_0 + R_0\alpha\Delta T + R_0 - R_0\alpha\Delta T$$

$$= 2R_0 \quad (I_A \text{ const})$$

Parallel:

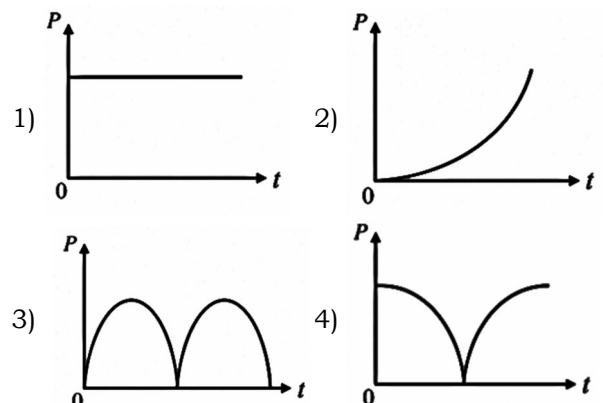
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R} = \frac{2}{R_0[1 - (\alpha\Delta T)^2]}$$

$$R = \frac{R_0[1 - (\alpha\Delta T)^2]}{2}$$

$$I_B = \frac{V}{R_{eqB}}; \quad I_B \text{ increases}$$

17. A conducting loop of finite resistance lies on the x-y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :



Ans. 4

Sol. $\phi = B \cdot A(t)$

$$= BA_0[1 + \sin t]$$

$$\text{emf} = -\frac{d\phi}{dt} = -BA_0 \left\{ \frac{d}{dt}[1 + \sin t] \right\} = -BA_0 \cos t$$

$$P = \frac{E^2}{R} = \frac{(-BA_0 \cos t)^2}{R} = \frac{B^2 A_0^2}{R} \cos^2 t$$

$P \propto \cos^2 t$ oscillation between maximum and minimum.

18. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon

$$p_{ph}, \frac{p_e}{p_{ph}} \text{ is}$$

(Take speed of light $= 3 \times 10^8 \text{ ms}^{-1}$, charge of electron $= 1.6 \times 10^{-19} \text{ C}$ and mass of electron $= 9 \times 10^{-31} \text{ kg}$)

- 1) 225 2) 275 3) $\frac{2}{450}$ 4) $\frac{1}{250}$

Ans. 1

Sol. $E = P \cdot c$

$$P = E / c$$

$$P_e = \sqrt{2mE}$$

$$\frac{P_e}{P_{ph}} = \frac{\sqrt{2mE} \times c}{E} = \sqrt{\frac{2m}{E}} \times c$$

$$E = 20 \text{ eV} = 3.2 \times 10^{-18}$$

$$= 3 \times 10^8 \sqrt{\frac{2 \times 9 \times 10^{-31}}{3.2 \times 10^{-18}}} = 225$$

19. Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-4} b$ is

- 1) $[L^{-1}T^{-1}K^{-3}]$ 2) $[L^{-1}T^{-1}K^{-4}]$
3) $[L^{-1}T^{-1}K^{-2}]$ 4) $[L^{-1}K^{-2}]$

Ans. 3

Sol. $\sigma_s \rightarrow M^1T^{-3}K^{-4}$

$$k_B \rightarrow M^1L^2T^{-2}K^{-1}$$

$$b \rightarrow L^1K^1$$

$$\frac{\sigma_s \cdot b}{k_B} = \frac{M^1T^{-3}K^{-4} \times L^1K^1}{M^1L^2T^{-2}K^{-1}} = L^{-1}T^{-1}K^{-2}$$

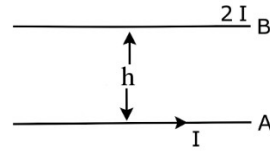
20. Two infinitely long parallel conducting wires A and B carry currents I and $2I$, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is :

[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

- 1) $\frac{2\mu_0 I^2}{\pi \lambda g}$ 2) $\frac{4\mu_0 I^2}{\pi \lambda g}$ 3) $\frac{\mu_0 I^2}{2\pi \lambda g}$ 4) $\frac{\mu_0 I^2}{\pi \lambda g}$

Ans. 4

Sol.



$$\lambda = \frac{m}{l}$$

$$F = mg$$

$$\frac{\mu_0 I_1 I_2 \cdot l}{2\pi(h)} = mg$$

$$\frac{\mu_0 (I)(2I)}{2\pi h} = \frac{m}{l} g$$

$$\frac{\mu_0 I^2}{\pi h} = \lambda g$$

$$h = \frac{\mu_0 I^2}{\pi \lambda g}$$

21. Which of the following measurements require 'index correction' ?

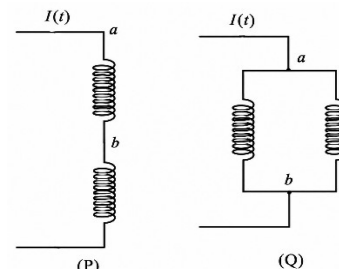
- 1) Measurement of focal length of lenses using optical bench
2) Measurement of speed of sound using resonance tube
3) Measurement of resistance of a wire using meter bridge
4) Measurement of gravitational acceleration using simple pendulum

Ans. 1

Sol. Index correction.

An adjustment made to distance measurement on an optical bench.

22. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q . The ratio E_P / E_Q is [Neglect the effect of mutual inductance.]

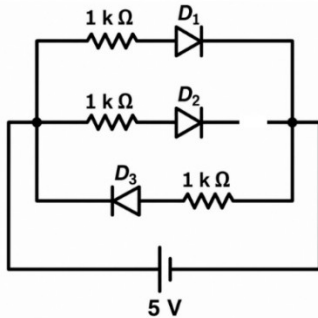


- 1) 1 2) 2 3) 1/4 4) 1/2

Ans. 2

Sol. $\varepsilon = -L \frac{dI(t)}{dt}$
 $\varepsilon \propto L$
 $\frac{\varepsilon_P}{\varepsilon_Q} = \frac{L}{L/2} = 2$

23. Three identical p-n junction diodes D_1 , D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1 , D_2 and D_3 are W_1 , W_2 and W_3 , respectively, then the correct option is :

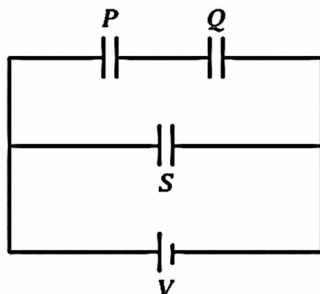


- 1) $W_3 > W_2 > W_1$ 2) $W_2 > W_1 = W_3$
 3) $W_1 > W_2 > W_3$ 4) $W_3 = W_1 > W_2$

Ans. 1

Sol. D_1 is forward biased
 D_2 is unbiased
 D_3 is reverse biased
 $W_3 > W_2 > W_1$

24. Three identical capacitors P, Q and S, each of the capacitance C , are connected to a battery of voltage V , as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T , respectively, then the ratio $\frac{U_P}{U_T}$ is :



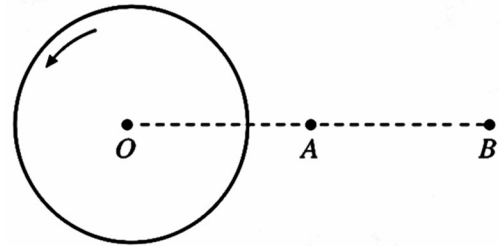
- 1) $1/2$ 2) $1/6$ 3) $2/3$ 4) $1/3$

Ans. 2

Sol. P, Q are in series $C_s = \frac{C}{2}$
 C_s, C are in parallel
 $\Rightarrow C_{eq} = \frac{C}{2} + C = \frac{3}{2}C$

$$\frac{U_P}{U_T} = \frac{\frac{1}{2}C\left(\frac{V}{2}\right)^2}{\frac{1}{2}\left(\frac{3C}{2}\right)V^2} = \frac{1}{6}$$

25. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is :

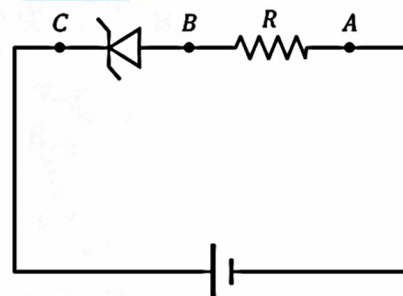


- 1) 1 2) 2 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

Ans. 1

Sol. $L = L_{cm} + mv_{cm}r$
 $v_{cm} = 0$
 $L = L_{cm}$
 $\frac{L_A}{L_B} = 1$

26. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5\text{ V}$. The magnitude of voltage difference between points B and A is :



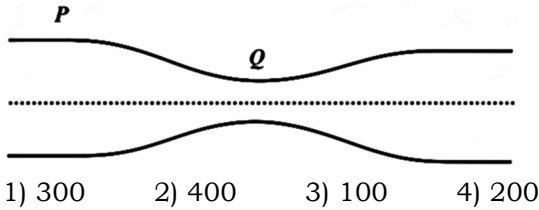
- 1) 1 V 2) 0 V 3) 3 V 4) 2 V

Ans. 4

Sol. Apply KVL
 $\therefore V_{BA} = 2\text{ V}$
 $\therefore V_{BA} = -2\text{ V}$
 Magnitude of potential difference $V_{BA} = 2\text{ V}$

27. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15 Nm^{-2} . The area

of cross-section at P and Q are 40 cm^2 and 20 cm^2 , respectively. The rate of flow of water through the pipe, in $\text{cm}^3 \text{ s}^{-1}$, is :
[Take density of water = 1000 kg m^{-3}]



Ans. 2

Sol. Apply equation of continuity

$$A_P v_P = A_Q v_Q$$

$$40 v_P = 20 v_Q$$

$$2 v_P = v_Q \quad \dots\dots\dots (1)$$

$$P_P - P_Q = \frac{1}{2} \rho (v_Q^2 - v_P^2)$$

$$15 = \frac{1}{2} \times 10^3 ((2v_P)^2 - v_P^2)$$

$$v_P = 10^{-1} \text{ ms}^{-1}$$

The rate of fluid flow

$$\frac{dV}{dt} = A_P v_P$$

$$= \frac{40}{10^{-2}} \times 10^{-1} = 400 \text{ cm}^3 / \text{s}$$

28. A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0 , α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is :

- 1) $\frac{1}{3}$ 2) $\frac{1}{2}$ 3) 3 4) 1

Ans. 2

Sol. $f_k(x) = \mu_k(x) \times N$

$$f_k(x) = (\mu_0 - \alpha x) Mg$$

When $\mu_k(L) = 0$

$$\mu_0 - \alpha L = 0$$

$$\Rightarrow \alpha = \frac{\mu_0}{L}$$

$$|W| = \int_0^L f_k(x) dx$$

$$\begin{aligned} &= \int_0^L (\mu_0 - \alpha x) Mg \\ &= Mg \frac{\mu_0 L}{2} \end{aligned}$$

$$\text{Comparing with } n\mu_0 MgL \Rightarrow n = \frac{1}{2}$$

29. A ray of light with wavelength λ is incident on three different photoelectric cells namely 1, 2 and 3. The threshold wavelength of these photoelectric cells are λ_1, λ_2 , and λ_3 , respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3 , respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda, \lambda_2 > \lambda$ and $\lambda_3 \gg \lambda$. The correct option is :

- 1) $V_1 > V_2, V_3 = 0$ 2) $V_1 < V_2, V_3 = 0$
3) $V_1 = 0, V_2 < V_3$ 4) $V_1 = 0, V_2 > V_3$

Ans. 3

Sol. In case I, $\lambda_1 < \lambda, V_1 = 0$

In case II, $\lambda_2 > \lambda, V_2 > 0$

In case III, $\lambda_3 \gg \lambda, V_3 > 0$

Since $\lambda_3 > \lambda_2$

According to $eV = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$, a large λ_0 results in a higher stopping potential.

$$\therefore V_2 < V_3$$

$$V_1 = 0 \text{ and } V_2 < V_3$$

30. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5 \text{ ms}^{-1}$. The radius of the orbit is $x \times 10^{-9} \text{ m}$. The value of x is :

[Take the mass of electron to be $9 \times 10^{-31} \text{ kg}$, charge of electron

$$= -1.6 \times 10^{-19} \text{ C and } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}]$$

- 1) 2 2) 1 3) 4 4) 3

Ans. 2

Sol. When an electron revolves in a circular orbit

$$F_c = F_e,$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r^2} \quad (\text{For hydrogen } Z = 1)$$

$$r = \frac{1}{4\pi\epsilon_0} \times \frac{e^2}{mv^2}$$

$$r = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{9 \times 10^{-31} \times 25.6 \times 10^{10}}$$

$$r = \frac{2.56 \times 10^{-29}}{25.6 \times 10^{-21}} = 0.1 \times 10^{-8} \text{ m}$$

$$r = 1 \times 10^{-9} \text{ m}$$

on comparing $x = 1$

31. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ . If the car travels at a speed 10 ms^{-1} , then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be 10 ms^{-2} , the value of θ is :

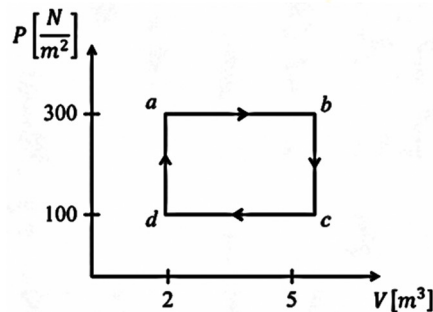
- 1) $\tan^{-1}(\sqrt{3}/2)$ 2) $\tan^{-1}(2\sqrt{3})$
3) $\tan^{-1}(\frac{1}{5})$ 4) $\tan^{-1}(\frac{2}{5})$

Ans. 3

Sol. $\tan \theta = \frac{v^2}{rg}$

$$\tan \theta = \frac{10 \times 10}{50 \times 10} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{5}\right)$$

32. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :



- 1) 600 J 2) 800 J 3) 400 J 4) 500 J

Ans. 1

Sol. 1st law of thermodynamics $\Delta Q = \Delta U + \Delta w$

Cyclic process, $\Delta U = 0$

Heat supplied = work

Work = area under P-V graph

$$= ad \times ab$$

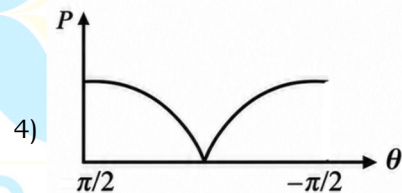
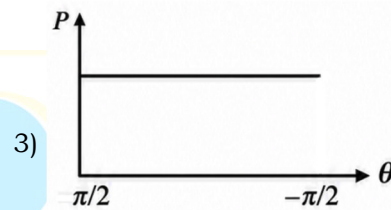
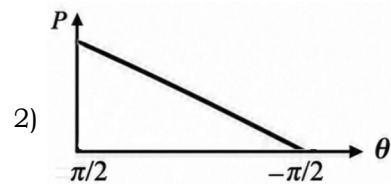
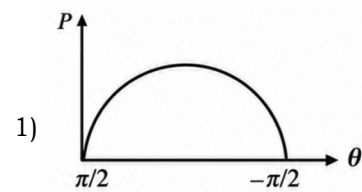
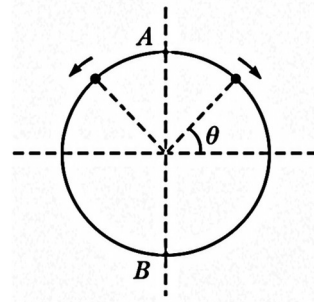
$$= (300 - 100) \times (5 - 2)$$

$$= 200 \times 3$$

$$\Delta Q = 600 \text{ J}$$

33. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point A $\left(\theta = \frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point B $\left(\theta = -\frac{\pi}{2}\right)$

During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?



Ans. 1

Sol. $y_1 = A \cos\left(\frac{\pi}{2} - \theta\right) = A \sin \theta$

$$v_1 = +\omega A \cos \theta$$

$$p_1 = mv_1 = m\omega A \cos \theta$$

$$y_2 = A \cos\left(\frac{\pi}{2} + \theta\right) = -A \sin \theta$$

$$v_2 = -A\omega \cos \theta$$

$$p_2 = -mA\omega \cos \theta$$

$$\text{Total momentum } \vec{p} = \vec{p}_1 + \vec{p}_2$$

$\vec{p}_1$ and $\vec{p}_2$ are opposite directions.

$$p = mA\omega \cos \theta - (-mA\omega \cos \theta)$$

$$p = 2mA\omega \cos \theta$$

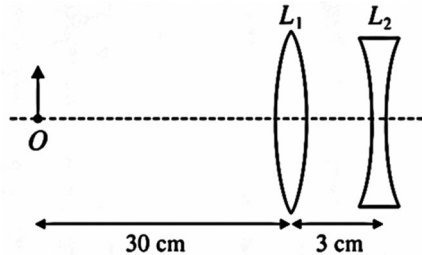
$$m = 1, A = 1, \omega = 1$$

$$\therefore p = 2 \cos \theta$$

Initial point A, $\theta = \frac{\pi}{2}$, $p = 0$

Final point B, $\theta = \frac{\pi}{2}$, $p = 0$

34. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2 , of the focal lengths +10 cm and -10 cm, respectively. The position of the image formed is :



- 1) 30 cm to the right of the concave lens
- 2) 60 cm to the right of the concave lens
- 3) 20 cm to the left of the concave lens
- 4) 60 cm to the left of the concave lens

Ans. 4

Sol. For L_1 :

$$\frac{1}{V_1} = \frac{1}{f_1} + \frac{1}{u_1} = \frac{1}{10} + \frac{1}{-30}$$

$$V_1 = +15 \text{ cm} \dots (1)$$

For L_2 :

$$\frac{1}{V_2} = \frac{1}{f_2} + \frac{1}{u_2} = \frac{1}{-10} + \frac{1}{12}$$

$$V_2 = -60 \text{ cm}$$

35. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2 , respectively, where $R_2 = \frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2 respectively. Then $\frac{v_2}{v_1}$ is :

- 1) $\sqrt{2}$
- 2) 2
- 3) $\frac{1}{\sqrt{2}}$
- 4) 1

Ans. 1

Sol. $V_e \propto \frac{1}{\sqrt{R}}$

$$\frac{V_2}{V_1} = \sqrt{\frac{R_1}{R_2}} = \sqrt{2}$$

36. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r = 9$, has the electric field, $E_x = E_0 \sin(kz - 2\pi \times 10^6 t) \text{ Vm}^{-1}$ where E_0 is the amplitude and k is the wave vector. Among

the following options, the incorrect choice is :

- 1) The magnetic field is given by the relation $B_y = \frac{B_0}{v} \sin(kz - 2\pi \times 10^6 t)$ where v is the speed of the electromagnetic wave inside the medium
- 2) The direction of propagation of the electromagnetic wave is along $+z$
- 3) The speed of the electromagnetic wave inside the medium is 10^8 ms^{-1}
- 4) The wavelength of the electromagnetic wave inside the medium is 300 m

Ans. 1 and 4

Sol. Step-I: $v = \frac{E_0}{B_0}$

$$B_0 = \frac{E_0}{v}$$

Step-II: Here $\mu_r = 1$

$$\text{Wave length } \lambda = \frac{v}{f} = \frac{10^8}{10^6} = 100 \text{ m}$$

37. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1 \text{ ms}^{-2}$, $\beta = 6 \text{ ms}^{-1}$ and $\gamma = 5 \text{ m}$. The average speed of the particle, in ms^{-1} , from $t = 0$ to $t = 6 \text{ s}$ is :

- 1) 3
- 2) 0
- 3) 12
- 4) 6

Ans. 1

Sol. $S(t) = t^2 - 6t + 5$

$$v(t) = \frac{ds}{dt}$$

$$v(t) = 2t - 6$$

$$\text{at } t = 3 \text{ sec, } v = 0$$

$$\therefore \text{Distance} = \left| \int_0^3 v dt \right| + \left| \int_3^6 v dt \right|$$

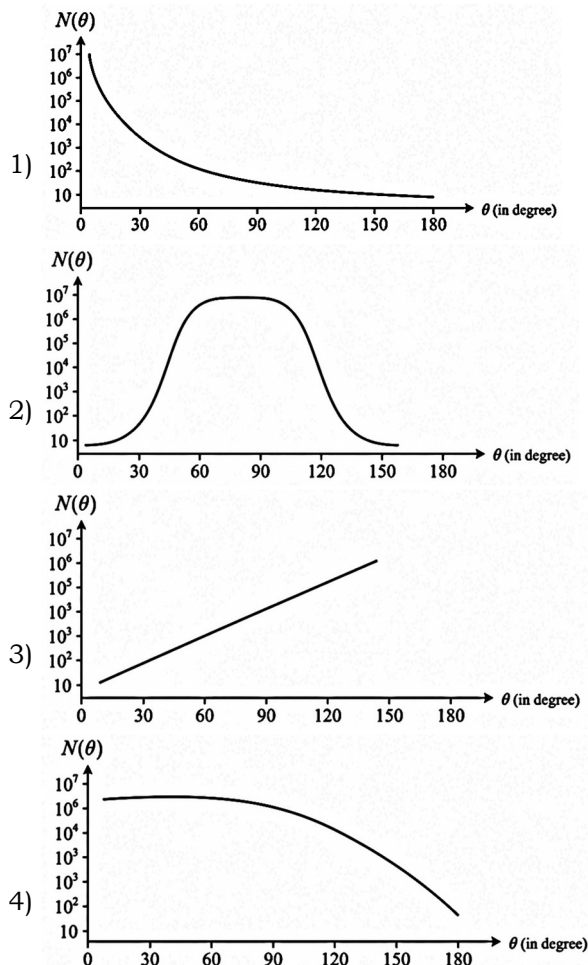
$$\left| \int_0^3 (2t - 6) dt \right| + \left| \int_3^6 (2t - 6) dt \right|$$

$$|9 + 9|$$

$$= 18$$

$$\text{Average speed} = \frac{\text{distance}}{\text{time}} = \frac{18}{6} = 3 \text{ m/sec}$$

38. In Geiger-Marsden experiment, the number of scattered α -particles $N(\theta)$ is plotted as a function of scattering angle θ . Which of the following options represents the correct plot?



Ans. 1

Sol. $N \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$

39. For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m , the ratio $\frac{n}{m}$ is :

- 1) 1 2) $\frac{3}{5}$ 3) $\frac{5}{9}$ 4) $\frac{9}{5}$

Ans. 1

Sol. No of nodes for 5th harmonic of an open pipe
 $n = 5$
 No of nodes for 9th harmonic of the closed pipe
 $m = 5$
 $\frac{n}{m} = \frac{5}{5}$

40. Consider a long solenoid of length l and radius r . If n is the number of turns per unit length

and μ_0 is the permeability of free space, the inductance of the solenoid is :

- 1) $(\mu_0 / 2\pi)n^2r^2l$ 2) $2\mu_0\pi n^2r^2l$
 3) $\mu_0\pi n^2r^2l$ 4) $\mu_0n^2r^2l$

Ans. 3

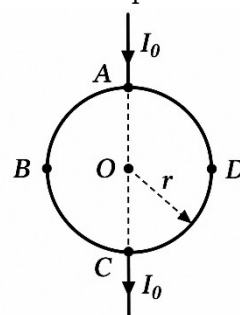
Sol. $LI = N\phi = (n\ell)(BA)$

For solenoid $B = \mu_0 nI$

$LI = (n\ell)(\mu_0 nI \times \pi r^2)$

$\Rightarrow L = \mu_0\pi n^2r^2\ell$

41. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :



- 1) $\frac{\mu_0 I_0}{2r}$ 2) $\frac{\mu_0 I_0}{2\pi r}$ 3) $\frac{\mu_0 I_0}{12r}$ 4) $\frac{\mu_0 I_0}{4r}$

Ans. 3

Sol. $I_{ABC} = I_0 \left(\frac{2R}{R + 2R} \right) = \frac{2I_0}{3}$

$I_{ADC} = I_0 \left(\frac{R}{R + 2R} \right) = \frac{I_0}{3}$

$B_{ABC} = \frac{\mu_0 I_{ABC}}{4r} = \frac{\mu_0 I_0}{6r}$

$B_{ADC} = \frac{\mu_0 I_{ADC}}{4r} = \frac{\mu_0 I_0}{12r}$

$B_{net} = B_{ABC} - B_{ADC} = \frac{\mu_0 I_0}{6r} - \frac{\mu_0 I_0}{12r}$

$B_{net} = \frac{\mu_0 I_0}{12r}$

42. The temperature of a metallic sphere of radius R is increased by a small amount ΔT . If the linear coefficient of thermal expansion of the metal is α , the approximate increase in the volume of the sphere is :

- 1) $4\pi R^3 \alpha \Delta T$ 2) $6\pi R^3 \alpha \Delta T$
 3) $2\pi R^3 \alpha \Delta T$ 4) $3\pi R^3 \alpha \Delta T$

Ans. 1

Sol. Increase in volume of sphere

$\Delta V = V\gamma \Delta T$

$$= \frac{4}{3} \pi R^3 (3\alpha) \Delta T$$

$$= 4\pi R^3 \alpha \Delta T$$

43. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of a wire to be 1 cm, the actual length of the wire is :

- 1) 1.00 cm 2) 1.04 cm
3) 0.60 cm 4) 0.96 cm

Ans. 2

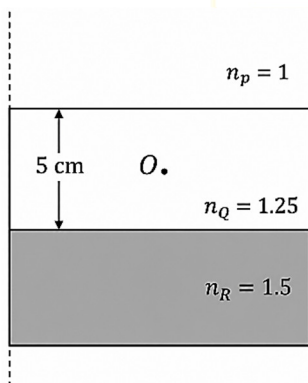
Sol. Least count $LC = \frac{1MSD}{n} = \frac{1mm}{10} = 0.01cm$

Error = $-4 \times 0.01 = -0.04cm$

Correction = $+0.04cm$

actual length of the wire = $1cm + 0.04cm$

44. Consider three media P, Q and R with refractive indices 1, 1.25 and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1 . For similar observation from medium R, the apparent depth is h_2 . The value of $|h_1 - h_2|$, in cm, is :



- 1) 2 2) 3 3) 0 4) 1

Ans. 4

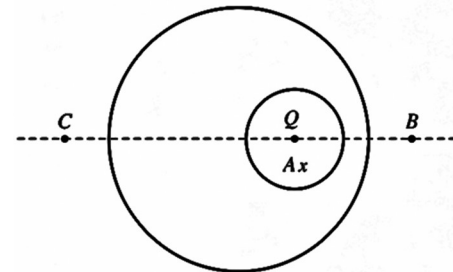
Sol. apparent depth = real depth / μ

$$h_1 = \frac{2.5}{1.25} = 2cm$$

$$h_2 = \frac{2.5}{\left(\frac{1.25}{1.5}\right)} = 3$$

$$h = |h_1 - h_2| = 1cm$$

45. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A , E_B and E_C , respectively. The points B and C are at the same distance from the center of the solid sphere.



- 1) $E_A = 0, E_B > E_C$ 2) $E_A \neq 0, E_B < E_C$
3) $E_A = 0, E_B = E_C$ 4) $E_A \neq 0, E_B = E_C$

Ans. 4

Sol. $E_A \neq 0$

$$E_B = E_C$$

CHEMISTRY QUESTION PAPER ANALYSIS

CHAPTER	NO. OF QUESTIONS
I PUC	
SOME BASIC CONCEPTS OF CHEMISTRY	2
STRUCTURE OF ATOM	1
CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES	2
CHEMICAL BONDING AND MOLECULAR STRUCTURE	3
THERMODYNAMICS	3
EQUILIBRIUM	1
REDOX REACTIONS	1
ORGANIC CHEMISTRY – SOME BASIC PRINCIPLES AND TECHNIQUES	4
HYDROCARBONS	2
II PUC	
SOLUTIONS	3
ELECTROCHEMISTRY	2
CHEMICAL KINETICS	2
P-BLOCK ELEMENTS	0
THE D-AND F-BLOCK ELEMENTS	4
COORDINATION COMPOUNDS	7
HALOALKANES AND HALOARENES	1
ALCOHOLS, PHENOLS AND ETHERS	-
ALDEHYDES, KETONES AND CARBOXYLIC ACIDS	3
AMINES	1
BIOMOLECULES	3
PRACTICAL CHEMISTRY	-

CHEMISTRY

46. The lanthanide ion having four unpaired electrons is

(Given: Atomic numbers of Ce = 58, Nd = 60, Tb = 65 and Ho = 67)

- 1) Tb³⁺ 2) Ho³⁺ 3) Nd³⁺ 4) Ce³⁺

Ans. 2

Sol. Ho³⁺ = [Xe] 4f¹⁰. It has 4 unpaired electrons

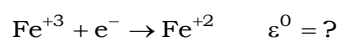
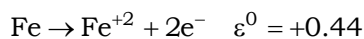
47. The standard electrode potential (E⁰) for the half-cell reaction Fe³⁺ + e⁻ → Fe²⁺ at 298 K is
(Given: E⁰(Fe³⁺ / Fe) = -0.04 V and

$$E^0(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V at } 298 \text{ K})$$

- 1) -0.48 V 2) +0.92 V
3) +0.40 V 4) +0.76 V

Ans. 4

Sol. Fe³⁺ + 3e⁻ → Fe ε⁰ = -0.04



$$n_3 E_3^0 = n_1 E_1^0 + n_2 E_2^0$$

$$1(E^0) = 3(-0.04) + 2(0.44)$$

$$E^0 = -0.12 + 0.88 = 0.76 \text{ V}$$

48. Given below are two statements:

Statement-I: Heating NaCl with concentrated H₂SO₄ and MnO₂ results in oxidation of Mn.

Statement-II: Heating NaI with concentrated H₂SO₄ and s MnO₂ results in reduction of Mn.

In light of the above statements, choose the *most appropriate* answer from the options given below:

- 1) Statement-I is correct but Statement-II is incorrect.
2) Statement-I is incorrect but Statement-II is correct.
3) Both Statement-I and Statement II are correct.
4) Both Statement-I and Statement II are incorrect.

Ans. 2

Sol. Conceptual

49. The complex which has facial and meridional isomers is

(Given: py = pyridine and en = H₂N -



- 1) [Co(NH₃)₄(H₂O)₂]³⁺ 2) [Ni(en)₂(H₂O)₂]²⁺
3) [Cr(py)₃(Cl)₃] 4) [Cr(H₂O)₆]³⁺

Ans. 3

Sol. Complexes with Ma₃b₃ shows facial and meridional isomers

50. Among the following, the compound having conjugated double bonds is

- 1) hepta-1,5-diene 2) hepta-1,6-diene
3) hepta-1,3-diene 4) hepta-1,4-diene

Ans. 3

Sol. H₃C - CH₂ - H₂C - HC = CH - CH = CH₂

51. Match the species in List I with their geometry in List II.

List I		List II	
A	PCl ₅	I	Tetrahedral
B	BrF ₅	II	Square Planar
C	BF ₄ ⁻	III	Trigonal bipyramidal
D	[Ni(CN) ₄] ²⁻	IV	Square pyramidal

Choose the correct answer from the options given below:

- 1) A-III, B-I, C-II, D-IV 2) A-III, B-II, C-I, D-IV
3) A-IV, B-III, C-I, D-II 4) A-III, B-IV, C-I, D-II

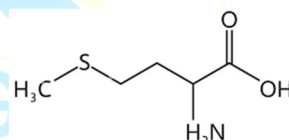
Ans. 4

Sol. Conceptual

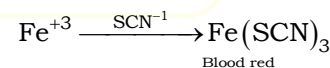
52. The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

- 1) methionine 2) serine
3) leucine 4) threonine

Ans. 1



Sol.



53. A protein undergoes reversible thermal denaturation from its initial state **N** to denatured state **D** according to **N** ⇌ **D**. At 60°C, the concentrations of both **N** and **D** are equal at equilibrium, and the standard enthalpy change of denaturation is 666 kJ mol⁻¹. The standard entropy change (ΔS⁰ in kJ K⁻¹ mol⁻¹) of the protein upon denaturation at 60°C is closest to

- 1) 333.0 2) 11.1 3) 2.0 4) 2000.0

Ans. 3

Sol. ΔH⁰ = TΔS⁰

$$\Delta S^0 = \Delta H^0 / T = \frac{666}{333} = 2$$

54. $2A \xrightarrow{k} B$ is a zero-order reaction, where $k = 1.0 \text{ mol L}^{-1} \text{ min}^{-1}$. If the initial concentration of A is 2 M, then the time taken to complete 75% of the reaction will be
- 1) 1.0 min
 - 2) 2.0 min
 - 3) 1.5 min
 - 4) 0.75 min

Ans. 3

Sol. $t = \frac{A_0 - A}{K} = \frac{2 - \frac{1}{2}}{1} = 1.5$

55. Given below are two statements:

Statement-I: $[\text{Fe}(\text{ox})_3]^{3-}$ is chiral.

Statement-II: $\text{trans} - [\text{Cr}(\text{H}_2\text{O})_2(\text{ox})_2]^-$ is chiral.

(Given: $\text{oxH}_2 = \text{HOOC} - \text{COOH}$)

In light of the above statements, choose the *most appropriate* answer from the options given below:

- 1) Statement-I is correct but Statement-II is incorrect.
- 2) Statement-I is incorrect but Statement-II is correct.
- 3) Both Statement-I and Statement-II are correct.
- 4) Both Statement-I and Statement-II are incorrect.

Ans. 1

Sol. $\text{trans} - [\text{Cr}(\text{H}_2\text{O})_2(\text{ox})_2]^-$ is optically inactive due to presence of plane of symmetry

56. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Generally, 3d transition metals have high melting points.

Reason R: Involvement of 3d-electrons in addition to 4s-electrons in the interatomic metallic bonding.

In light of the above statements, choose the *most appropriate* answer from the options given below:

- 1) **A** is correct but **R** is not correct.
- 2) **A** is not correct but **R** is correct.
- 3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
- 4) Both **A** and **R** are correct and **R** is NOT the correct explanation of **A**.

Ans. 3

Sol. Involvement of 3d-electrons in addition to 4s-electrons in the interatomic metallic bonding the melting point of 3d-transition metals increases

57. Among the species given below, the spin-only magnetic moment is highest for (Given: Atomic number of Ti = 22, Mn = 25, Fe = 26 and Co = 27)

- 1) $[\text{Co}(\text{NH}_3)_6]^{3+}$
- 2) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
- 3) $[\text{Mn}(\text{CN})_6]^{3-}$
- 4) $[\text{Fe}(\text{CN})_6]^{3-}$

Ans. 3

Sol.

- 1) $\text{CO} = [\text{Ar}]4s^23d^7$ $\text{CO}^{+3} = [\text{Ar}]4s^03d^6$ No of unpaired $e^- = 0$
- 2) $\text{Ti} = [\text{Ar}]4s^23d^2$ $\text{Ti}^{+3} = [\text{Ar}]4s^03d^1$ No of unpaired $e^- = 1$
- 3) $\text{Mn} = [\text{Ar}]4s^23d^5$ $\text{Mn}^{+3} = [\text{Ar}]4s^03d^4$ No of unpaired $e^- = 2$
- 4) $\text{Fe} = [\text{Ar}]4s^23d^6$ $\text{Fe}^{+3} = [\text{Ar}]4s^03d^5$ No of unpaired $e^- = 1$

58. According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is

- 1) $\text{Cl}^- > \text{H}_2\text{O} > \text{NH}_3 > \text{CO}$
- 2) $\text{Cl}^- > \text{NH}_3 > \text{H}_2\text{O} > \text{CO}$
- 3) $\text{CO} > \text{NH}_3 > \text{H}_2\text{O} > \text{Cl}^-$
- 4) $\text{CO} > \text{H}_2\text{O} > \text{NH}_3 > \text{Cl}^-$

Ans. 3

Sol. $\text{CO} > \text{NH}_3 > \text{H}_2\text{O} > \text{Cl}^-$

59. In potash alum, the ratio of K^+ and SO_4^{2-} ions is

- 1) 2 : 3
- 2) 3 : 2
- 3) 1 : 2
- 4) 2 : 1

Ans. 3

Sol. $\frac{\text{K}^+}{\text{SO}_4^{2-}} = \frac{2}{4} = \frac{1}{2}$

60. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The first ionization enthalpy of O is lower than that of N and F.

Reason R: The loss of an electron from O leads to stable half-filled p orbital. In light of the above statements, choose the *most appropriate* answer from the options given below:

- 1) **A** is correct but **R** is not correct.
- 2) **A** is not correct but **R** is correct.
- 3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
- 4) Both **A** and **R** are correct and **R** is NOT the correct explanation of **A**.

Ans. 4

Sol. Conceptual

61. Given below are two statements!

Statement-I : Oxidation of p-nitrotoluene with acidic KMnO_4 gives an acid that is stronger than benzoic acid.

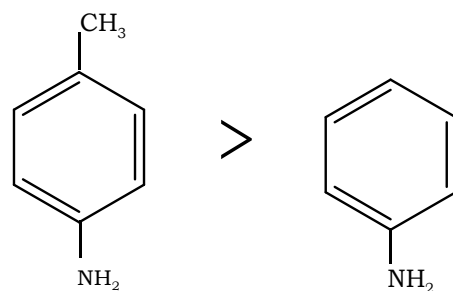
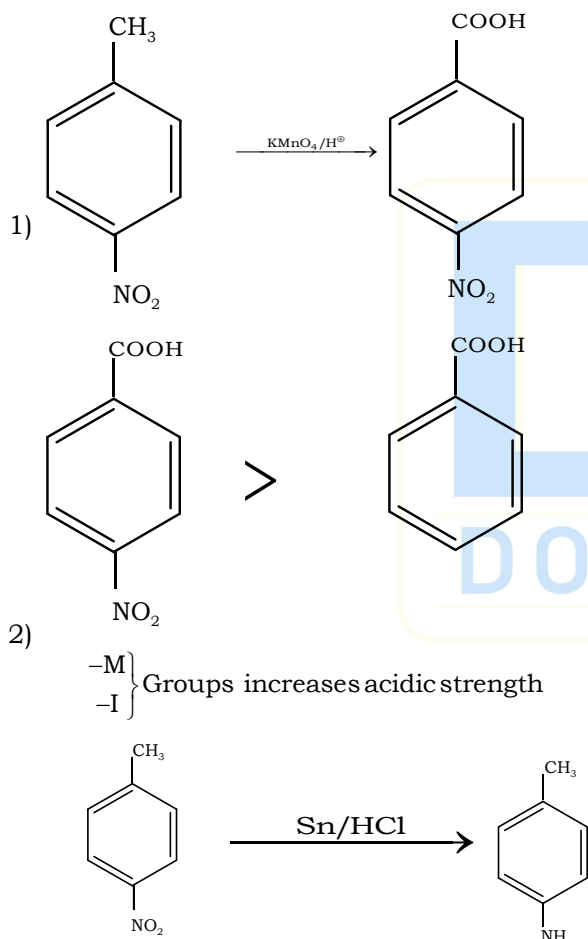
Statement-II : Reduction of p-nitrotoluene with Sn/HCl followed by neutralization gives an amine that is more basic than aniline.

In light of the above statements, choose the **most appropriate** answer from the options given below.

- 1) **Statement-I** is correct but **Statement-II** is incorrect.
- 2) **Statement-I** is incorrect but **Statement-II** is correct.
- 3) Both **Statement-I** and **Statement-II** are correct.
- 4) Both **Statement-I** and **Statement-II** are incorrect.

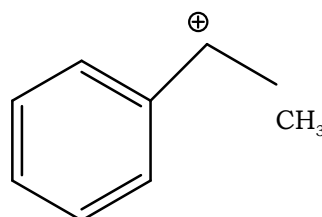
Ans. 3

Sol.



$-\text{CH}_3 \rightarrow \begin{matrix} +M \\ +I \end{matrix} \left. \vphantom{\begin{matrix} +M \\ +I \end{matrix}} \right\} \text{Groups Increases basic strength}$

62. The following carbocation is stabilized by the interaction of the empty p orbital with

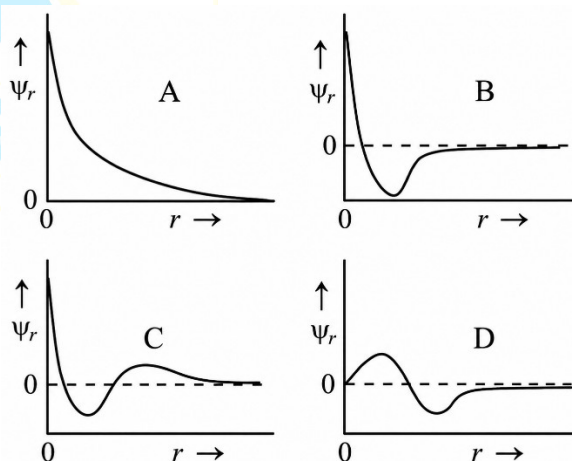


- 1) Empty σ^* and filled π orbitals
- 2) Empty σ^* and filled π^* orbitals
- 3) Filled σ and π orbitals
- 4) Empty σ and empty π^* orbitals

Ans. 3

Sol. Due to sigma from CH_3 & π from benzene

63. Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.

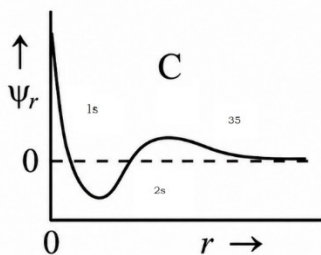


The figure representing two radial nodes in the orbital is

- 1) C
- 2) D
- 3) A
- 4) B

Ans. 1

Sol.



$$\begin{aligned}\text{Radial mode} &= n - \ell - 1 \\ &= 3 - 0 - 1 = 2\end{aligned}$$

In the plots of 3s orbital curve crosses the r(radius) axis exactly 2 lines resulting in 3 distinct pearls

64. The correct order of solubility of the given salts in water at 298K is

Salt	K_{sp} at 298K
AgBr	5×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

- 1) $Hg_2Cl_2 > AgBr > Zn(OH)_2$
- 2) $Zn(OH)_2 > AgBr > Hg_2Cl_2$
- 3) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$
- 4) $AgBr > Zn(OH)_2 > Hg_2Cl_2$

Ans. 2

Sol.
$$\begin{array}{ccc} Zn(OH)_2 & > & AgBr > & Hg_2Cl_2 \\ 6.3 \times 10^{-6} & & 7.07 \times 10^{-7} & 6.8 \times 10^{-7} \end{array}$$

65. A 1:3 electrolyte in an aqueous solution is

- 1) $[Co(NH_3)_6]Cl_3$
- 2) $[Co(NH_3)_3(NO_2)_3]$
- 3) $[CoCl_2(NH_3)_4]Cl$
- 4) $[CoCl(NH_3)_5]Cl_2$

Ans. 1

Sol. $[Co(NH_3)_6]Cl_3$ dissociates into one $[Co(NH_3)_6]^{3+}$, complex ions and $3Cl^-$ ions

Ratio: 1:3

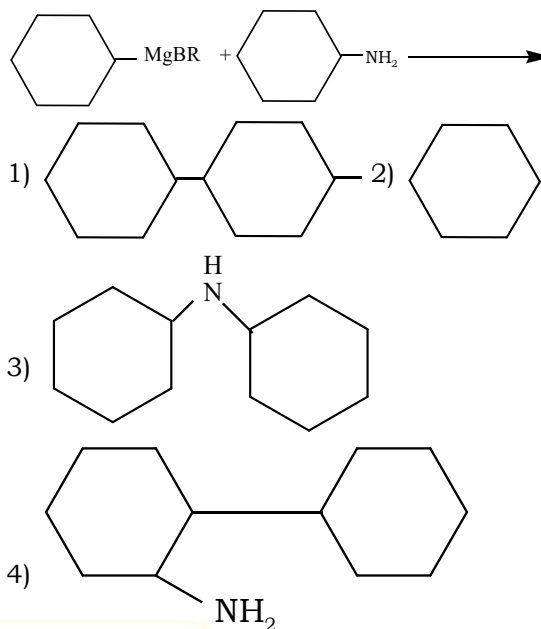
66. The correct statement about peptides and proteins is

- 1) In β -pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds
- 2) In α -helices, the polypeptide chain is twisted into a left-handed screw (helix) through intramolecular hydrogen bonds.
- 3) Tertiary structure of proteins has two or more polypeptide subunits.
- 4) Only the proteins having a quaternary structure are biologically active.

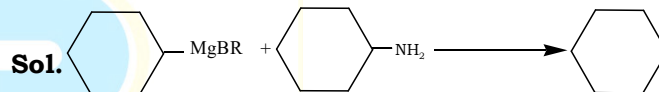
Ans. 1

Sol. β -pleated sheet structures chains are aligned side by side and these are held together by intermolecular H-bonds between the carbonyl oxygen of one chain and the amino hydrogen of the adjacent chain.

67. One of the products formed in the following reaction is



Ans. 2

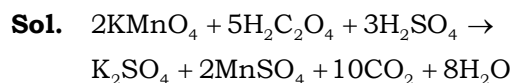


Protonation of alkyl group present in the Grignard's reagent takes place

68. In acidic medium, 10 mL of 0.25M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10mL, the strength of the $KMnO_4$ solution is

- 1) 0.25M
- 2) 0.15M
- 3) 0.10M
- 4) 0.20M

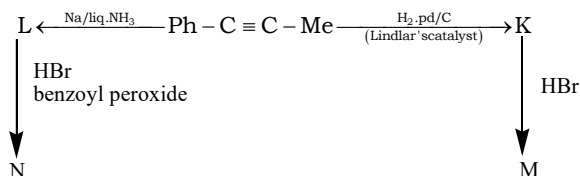
Ans. 3



$$\begin{aligned}\frac{M_1 V_1}{n_1} &= \frac{M_2 V_2}{n_2} \\ \frac{M_1 \times 10}{2} &= \frac{0.25 \times 10}{5}\end{aligned}$$

$$M_{(KMnO_4)} = 0.1M$$

69. Consider the following reaction sequences and choose the correct option.



- 1) M and N are geometrical isomers
- 2) M and N are stereoisomers
- 3) K and L are geometrical isomers
- 4) K and L are enantiomers

Ans. 3

Sol. K gives cis alkenes

L gives trans alkenes

Cis alkenes and trans alkenes are geometrical isomers

70. Arrange the following compounds in the increasing order of polarity

- A) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ B) $\text{CH}_3\text{CH}_2\text{OH}$
C) CH_3COCH_3 D) CH_3COOH

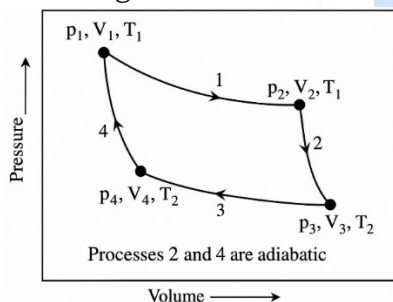
Choose the correct answer from the given below.

- 1) $\text{C} < \text{A} < \text{B} < \text{D}$ 2) $\text{A} < \text{C} < \text{B} < \text{D}$
- 3) $\text{A} < \text{B} < \text{C} < \text{D}$ 4) $\text{C} < \text{A} < \text{D} < \text{B}$

Ans. 2

Sol. Polarity order: Ethers < Ketones < Alcohols < carboxylic acids

71. Consider the reversible process for 1.0mol of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the process 1, 2, 3 and 4, respectively. ΔU_2 and ΔU_4 are changes in the internal energy for the process 2 and 4, respectively. [Use $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$]. The correct options is

- 1) $w_1 + w_2 = 2T_1 \ln \frac{V_2}{V_1}$
- 2) $w_1 + w_2 + w_3 + w_4 = 0$
- 3) $w_1 + w_3 = -2T_1 \ln \frac{V_2}{V_1} - 2T_2 \ln \frac{V_4}{V_3}$
- 4) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

Ans. 3

Sol. Process (1) (1-2) isothermal expansion
($T = T_1$)

$$w_1 = -nRT_1 \ln \left(\frac{V_2}{V_1} \right) = -2T_1 \ln \left(\frac{V_2}{V_1} \right)$$

Process (2) (2-3) adiabatic expansion ($Q=0$)

$$\Delta U_2 = w_2; \Delta U = nC_v \Delta T$$

$$w_2 = C_v (T_2 - T_1)$$

Process (3) (3-4) Isothermal compression

$$(T = T_2)$$

$$w_3 = -nRT_2 \ln \left(\frac{V_4}{V_3} \right) = -2T_2 \ln \left(\frac{V_4}{V_3} \right)$$

Process (4) (4-1) adiabatic compression ($Q=0$)

$$\Delta U_4 = w_4 = C_v (T_1 - T_2)$$

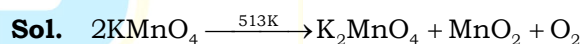
$$= -2T_1 \ln \left(\frac{V_2}{V_1} \right) - 2T_2 \ln \left(\frac{V_4}{V_3} \right)$$

$$\text{Hence } w_1 + w_3 = -2T_1 \ln \left(\frac{V_2}{V_1} \right) - 2T_2 \ln \left(\frac{V_4}{V_3} \right)$$

72. The green paramagnetic species formed by heating KMnO_4 at 513 K is

- 1) MnO 2) KO_2
- 3) K_2MnO_4 4) Mn_3O_4

Ans. 3



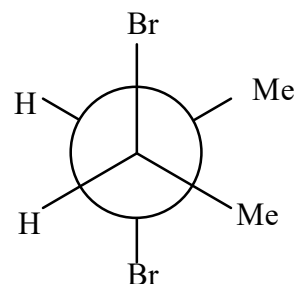
73. The number 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are
- 1) 17.014 and 21.024
 - 2) 17.015 and 21.024
 - 3) 17.014 and 21.023
 - 4) 17.015 and 21.023

Ans. 1

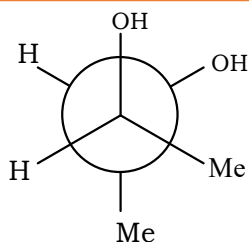
Sol. 17.014 and 21.024

74. Given below are two statements

Statement-I: trans-but-2-ene upon treatment with Br_2 in CCl_4 gives the following product.



Statement-II: cis-but-2-ene upon treatment with alkaline KMnO_4 gives the following product.

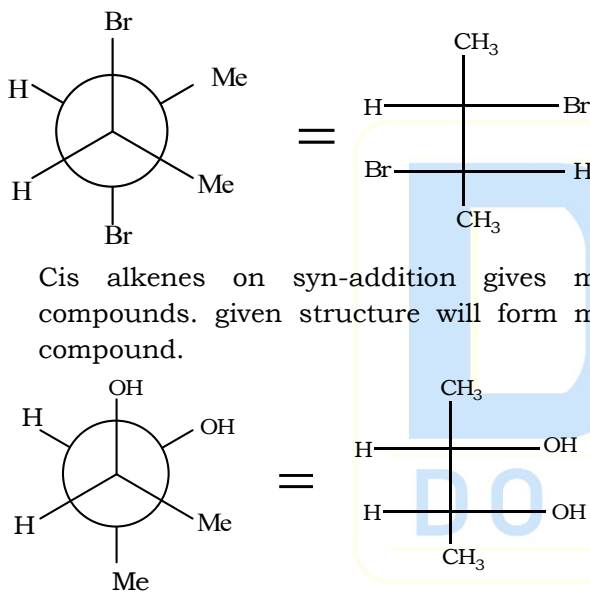


In the light of the above statements, choose the most appropriate answer from the options given below.

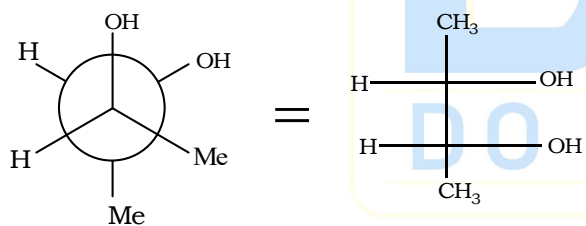
- 1) **Statement I** is correct but **Statement II** is incorrect
- 2) **Statement I** is incorrect but **Statement II** is correct
- 3) Both **Statement I** and **Statement II** are correct
- 4) Both **Statement I** and **Statement II** are incorrect

Ans. 2

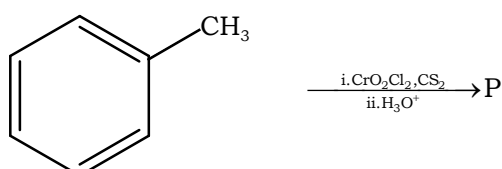
Sol. Trans alkenes on anti-addition gives meso compounds but given structure cannot form meso compound.



Cis alkenes on syn-addition gives meso compounds. given structure will form meso compound.



75. Consider the following reaction and choose the correct option.

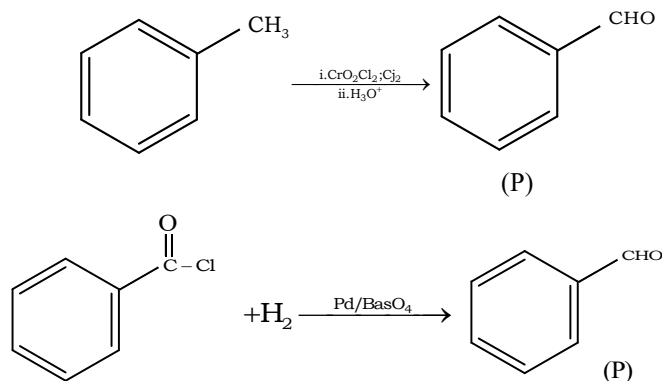


- 1) On treatment with bromine water, compound P gives a white precipitate
- 2) Compound P is obtained by the hydrogenation of benzoyl chloride with Pd on BaSO₄
- 3) On treating compound P with saturated NaHCO₃ solution, brisk effervescence is observed

- 4) Compound P can be prepared by treating benzene with anhydrous AlCl₃ and CH₃COCl

Ans. 2

Sol.



76. Match the vitamins in List I with their sources in List II

List I		List II	
A.	vitamin A	I.	meat
B.	vitamin B ₁₂	II.	sunflower oil
C.	vitamin E	III.	green leafy vegetables
D.	vitamin K	IV.	carrots

Choose the correct answer from the options given below.

- 1) A-IV, B-II, C-I, D-III
- 2) A-III, B-I, C-IV, D-II
- 3) A-II, B-III, C-IV, D-I
- 4) A-IV, B-I, C-II, D-III

Ans. 4

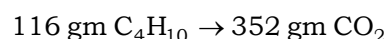
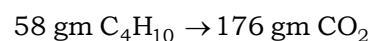
Sol. Conceptual

77. The amount of carbon dioxide evolved upon complete combustion of 116 g of n-butane is (Given: atomic mass in amu H = 1, C = 12 and O = 16)

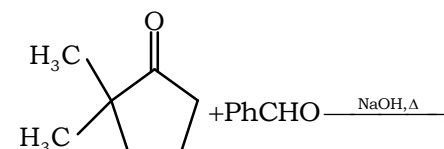
- 1) 176 g 2) 362 g 3) 352 g 4) 322 g

Ans. 3

Sol. $C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$



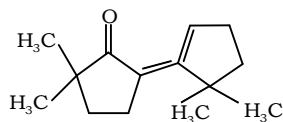
78. The compound that CANNOT be obtained from the aldol condensation reaction shown below, is



- 1)
- 2)
- 3)
- 4)

Ans. 3

Sol.

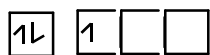


Cannot be formed by aldol condensation.

79. The correct statement is
- 1) Magnesium has covalency of four. a maximum
 - 2) Aluminium has five valence orbitals.
 - 3) Boron has a maximum covalency of four.
 - 4) Beryllium has three valence orbitals.

Ans. 3

Sol. $B - 5 - 1s^2 2s^2 2p^1$



Maximum covalency -4

80. The formula of tetraammineaquachloridocobalt (III) chloride is
- 1) $[Co(NH_3)_4(H_2O)Cl]Cl$
 - 2) $[Co(NH_3)_4(H_2O)Cl]Cl_2$
 - 3) $[Co(NH_3)_4Cl_2] \times H_2O$
 - 4) $[Co(NH_3)_4]Cl_3 \times H_2O$

Ans. 2

Sol. $[Co(NH_3)_4(H_2O)Cl]Cl_2$

81. **Assertion A:**

For an ideal solution formed by mixing liquids P and Q, $\Delta_{mix}H = 0$ and $\Delta_{mix}V = 0$.

Reason R:

No interactions occur between P and Q

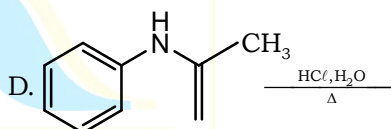
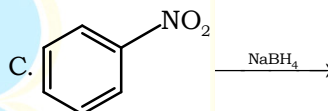
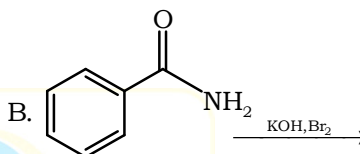
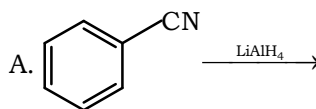
In the light of the above statements, choose the most appropriate answer from the options given below.

- 1) A is correct but R is not correct
- 2) A is not correct but R is correct
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are correct but R is NOT the correct explanation of A

Ans. 1

Sol. **Conceptual**

82. Identify the reactions which give aniline as the major product.



Choose the correct answer from the options given below.

- 1) A and C only
- 2) C and D only
- 3) A and B only
- 4) B and D only

Ans. 4

Sol. **Conceptual**

83. The correct decreasing order of oxidation state of the underlined atom in each molecule is
- 1) $PbO_2 > \underline{N}_2O_3 > \underline{S}O_3$
 - 2) $\underline{P}_4O_6 > \underline{Cl}_2O_7 > \underline{Al}H_3$
 - 3) $\underline{P}_4O_{10} > \underline{S}O_3 > \underline{H}_2O$
 - 4) $\underline{N}_2O_5 > \underline{Al}_2O_3 > \underline{H}_2S$

Ans. 4

Sol. $\underline{N}_2O_5 > \underline{Al}_2O_3 > \underline{H}_2S$

$N = +5, Al = +3, S = -2$

89. For a salt **XY**, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of $-90.0 \text{ Scm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$ at 298 K. At 0.01 M concentration of **XY**, the value of Λ_m is $145.0 \text{ Scm}^2 \text{ mol}^{-1}$. The limiting molar conductivity of **Y**-ion ($\lambda_{Y^-}^0$, in $\text{Scm}^2 \text{ mol}^{-1}$) at 298 K will be
(Given: $\lambda_{X^+}^0 = 74.0 \text{ Scm}^2 \text{ mol}^{-1}$)
- 1) 90.0 2) 76.0 3) 80.0 4) 100.0

Ans. 3

Sol. $\lambda_m = \lambda_m^0 - A\sqrt{C}$

$$145 = \lambda_m^0 - 90\sqrt{10^{-2}}$$

$$145 = \lambda_m^0 - 90(10^{-1})$$

$$\lambda_m^0 = 145 + 9 = 154$$

$$\Rightarrow \lambda_{m(xy)}^0 = \lambda_{m(X^+)}^0 + \lambda_{m(Y^-)}^0$$

$$154 = 74 + \lambda_{m(Y^-)}^0$$

$$\lambda_{m(Y^-)}^0 = 80 \text{ Scm}^2 \text{ mol}^{-1}$$

90. Among the following options, the correct trend in the electron gain enthalpy is
- 1) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ 2) $\text{I} > \text{Br} > \text{Cl} < \text{F}$
3) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ 4) $\text{Br} > \text{Cl} > \text{F} > \text{I}$

Ans. 1

Sol. Conceptual

BIOLOGY QUESTION PAPER ANALYSIS

CHAPTER

NO. OF QUESTIONS

I PUC

THE LIVING WORLD

3

BIOLOGICAL CLASSIFICATION

4

PLANT KINGDOM

2

MORPHOLOGY OF FLOWERING PLANTS

3

ANATOMY OF FLOWERING PLANTS

-

CELL : THE UNIT OF LIFE

7

BIOMOLECULES

2

CELL CYCLE AND CELL DIVISION

2

PHOTOSYNTHESIS IN HIGHER PLANTS

5

RESPIRATION IN PLANTS

1

PLANT GROWTH AND DEVELOPMENT

3

II PUC

SEXUAL REPRODUCTION IN FLOWERING PLANTS

4

PRINCIPLES OF INHERITANCE AND VARIATION

1

MOLECULAR BASIS OF INHERITANCE

2

MICROBES IN HUMAN WELFARE

1

BIOTECHNOLOGY : PRINCIPLES AND PROCESSES

4

BIOTECHNOLOGY AND ITS APPLICATIONS

1

ECOSYSTEM

2

BIOLOGY QUESTION PAPER ANALYSIS

CHAPTER

NO. OF QUESTIONS

I PUC

ANIMAL KINGDOM

3

STRUCTURAL ORGANISATION IN ANIMALS

2

BREATHING AND EXCHANGE OF GASES

1

BODY FLUIDS AND CIRCULATION

4

EXCRETORY PRODUCTS AND THEIR ELIMINATION

1

LOCOMOTION AND MOVEMENT

5

NEURAL CONTROL AND COORDINATION

1

CHEMICAL COORDINATION AND INTEGRATION

3

II PUC

HUMAN REPRODUCTION

5

REPRODUCTIVE HEALTH

2

EVOLUTION

6

HUMAN HEALTH AND DISEASE

3

ORGANISMS AND POPULATIONS

5

BIODIVERSITY AND CONSERVATION

1

GENETIC DISORDERS (HUMAN GENETICS)

1

BIOLOGY

91. Given below are two statements :

Statement I: Chromosomes are fully condensed at the end of prophase I.

Statement II : Meiosis I resembles mitosis.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) **Statement I** is correct, but **Statement II** is false
- 2) **Statement I** is incorrect, but **Statement II** is true
- 3) Both **Statement I** and **Statement II** are true
- 4) Both **Statement I** and **Statement II** are false

Ans. 1

92. Which of the following is **not** a characteristic of chordates ?

- 1) Absence of gills
- 2) Presence of post anal part (tail)
- 3) Presence of notochord
- 4) Central nervous system is dorsal

Ans. 1

93. Phyllotaxy is the pattern of arrangement of _ .

- 1) fruits 2) sepals 3) leaves 4) flowers

Ans. 3

94. Which one of the following statements is **incorrect**?

- 1) Glucagon stimulates glycogenolysis
- 2) β -cells of pancreas secrete insulin
- 3) α -cells of pancreas secrete glucagon
- 4) α -cells of pancreas secrete insulin

Ans. 4

95. The plastid that stores xanthophyll is known as _____ .

- 1) aleuroplast 2) amyloplast
- 3) chloroplast 4) chromoplast

Ans. 4

96. Match **List-I** with **List-II**.

	List-I		List-II
A)	Fusion of protoplasts between gametes	I)	Meiosis
B)	Fusion of two nuclei	II)	Plasmogamy
C)	Generation of haploid spores	III)	Karyogamy

Choose the **correct** answer from the options given below :

- 1) A-III, B-II, C-I 2) A-I, B-III, C-II
- 3) A-II, B-III, C-I 4) A-II, B-I, C-III

Ans. 3

97. Given below are two statements :

Statement I: In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II : In gymnosperms, seeds are not covered.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) **Statement I** is correct but **Statement II** is incorrect
- 2) **Statement I** is incorrect but **Statement II** is correct
- 3) Both **Statement I** and **Statement II** are correct
- 4) Both **Statement I** and **Statement II** are incorrect

Ans. 3

98. In water, frogs respire using _____ .

- 1) lungs 2) trachea
- 3) skin 4) buccal cavity

Ans. 3

99. Mad cow disease is caused by _____ .

- 1) *Aspergillus sp.* 2) *Mycoplasma sp.*
- 3) prions 4) viroids

Ans. 3

100. Which of the following statements regarding photorespiration are correct ?

- (a) Do not occur in C3 plants,
- (b) CO_2 is consumed and O_2 is generated
- (c) Phosphoglycolate is formed
- (d) No synthesis of ATP and NADPH

Choose the **correct** answer from the options given below :

- 1) (b) and (d) only 2) (a) and (b) only
- 3) (a) and (d) only 4) (c) and (d) only

Ans. 4

101. Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.

- 1) Release of oxygen → production of glucose → absorption spectra of chlorophyll a and b → role of air
- 2) Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b
- 3) Absorption spectra of chlorophyll a and b → production of glucose → release of oxygen → role of air
- 4) Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b

Ans. 4

102. How many turns of Calvin cycle are required for the formation of three molecules of glucose?

- 1) 1 2) 18 3) 6 4) 3

Ans. 2

103. Mitochondrial inner membrane encloses ____.

- 1) mucus 2) aqueous humor
- 3) matrix 4) cytosol

Ans. 3

104. Which of the following statements is **incorrect**?

- 1) Fibrin is produced from fibrinogen
- 2) Fibrinogen is produced from fibrin
- 3) Blood coagulates in response to an injury
- 4) Blood clot consists of fibrins

Ans. 2

105. Sphenopsida class belongs to ____.

- 1) gymnosperms 2) pteridophytes
- 3) bryophytes 4) angiosperms

Ans. 2

106. Match **List-I** with **List-II**.

	List-I		List-II
A)	Spherical	I)	Vibrio
B)	Rod	II)	Cocci
C)	Comma	III)	Spirilla
D)	Spirillum	IV)	Bacilli

Choose the **correct** answer from the options given below :

- 1) A-II, B-I, C-IV, D-III 2) A-II, B-IV, C-I, D-III
- 3) A-I, B-III, C-II, D-IV 4) A-III, B-II, C-I, D-IV

Ans. 2

107. Which of the following are characteristics of prokaryotic cells ?

- (a) Ribosomes are made of 50S and 30S subunits
- (b) They can have plasmids
- (c) They contain mesosome
- (d) They have peroxisomes

Choose the **correct** answer from the options given below :

- 1) (a), (c) and (d) only 2) (a), (b) and (c) only
- 3) (b) and (c) only 4) (a) and (c) only

Ans. 2

108. Smooth endoplasmic reticulum ____.

- 1) is actively involved in protein synthesis
- 2) is a site for the synthesis of carbohydrates
- 3) has ribosomes attached to its surface
- 4) is the major site for the synthesis of lipids

Ans. 4

109. Given below are two statements :

Statement I : The class name Reptilia refers to creeping or crawling mode of locomotion

Statement II : All organisms belonging to Reptilia have three chambered heart.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) **Statement I** is correct but **Statement II** is incorrect
- 2) **Statement I** is incorrect but **Statement II** is correct
- 3) Both **Statement I** and **Statement II** are correct
- 4) Both **Statement I** and **Statement II** are incorrect

Ans. 1

110. In frogs, the number of pairs of cranial nerves arising from the brain are ____.

- 1) 10 2) 12 3) 6 4) 9

Ans. 1

111. Cell theory was formulated by ____.

- 1) Singer and Nicolson
- 2) Antonie Von Leeuwenhoek
- 3) Schleiden and Schwann
- 4) Robert Brown

Ans. 3

112. Which of the following statements related to pituitary gland are **correct** ?

- (a) It is divided anatomically into adenohypophysis and neurohypophysis
(b) It secretes follicle stimulating hormone
(c) It secretes melanocyte stimulating hormone
(d) It does not secrete prolactin

Choose the **correct** answer from the options given below :

- 1) (c) and (d) only 2) (b) and (c) only
3) (a) and (b) only 4) (a), (b) and (c) only

Ans. 4

113. Which of the following is **not** a prokaryote ?

- 1) Mycoplasma 2) Fungi
3) Bacteria 4) Blue green algae

Ans. 2

114. Length of the stem at time 0 is 20 cm. The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7th day ?

- 1) 230 cm 2) 460 cm 3) 50 cm 4) 170 cm

Ans. 1

115. Which of the following is **not** a part of human central neural system ?

- 1) Pia mater 2) Pericardium
3) Arachnoid 4) Dura mater

Ans. 2

116. Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage ?

- 1) Indole butyric acid 2) Ethephon
3) Absciscic acid 4) Gibberellin

Ans. 4

117. Match **List-I** with **List-II**.

	List-I		List-II
A)	Marginal placentation	I)	<i>Argemone</i>
B)	Axile placentation	II)	Tomato
C)	Parietal placentation	III)	<i>Primrose</i>
D)	Free central placentation	IV)	Pea

Choose the **correct** answer from the options given below :

- 1) A-IV, B-III, C-I, D-II 2) A-IV, B-II, C-I, D-III
3) A-II, B-IV, C-I, D-III 4) A-IV, B-II, C-III, D-I

Ans. 2

118. Which of the following plant growth regulators is used as herbicide ?

- 1) Absciscic acid 2) Gibberellin
3) 2,4-D 4) Kinetin

Ans. 3

119. Given below are two statements:

Statement I: When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.

Statement II: In phylum Echinodermata, both adults and larvae are radially symmetrical.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- 1) **Statement I** is correct but **Statement II** is incorrect
2) **Statement I** is incorrect but **Statement II** is correct
3) Both **Statement I** and **Statement II** are correct
4) Both **Statement I** and **Statement II** are incorrect

Ans. 1

120. Match **List-I** with **List-II**.

	List-I		List-II
A)	Starch	I)	Fights infection
B)	Antibody	II)	Energy storage
C)	Concanavalin A	III)	Glucose transport
D)	Glut-4	IV)	Lectin

Choose the **correct** answer from the options given below:

- 1) A-II, B-I, C-III, D-IV 2) A-I, B-II, C-III, D-IV
3) A-I, B-II, C-IV, D-III 4) A-II, B-I, C-IV, D-III

Ans. 4

121. The number of vertebrae in a human is ____.

- 1) 26 2) 206 3) 7 4) 12

Ans. 1

122. Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700)?

- 1) Xanthophylls 2) Carotenoids
3) Chlorophyll b 4) Chlorophyll a

Ans. 4

123. Arrange the following taxonomic categories in ascending order.

- (a) Genus (b) Class
(c) Order (d) Phylum
(e) Family (f) Kingdom
(g) Species

Choose the correct answer from the options given below:

- 1) (g), (c), (d), (b), (e), (a), (f)
2) (f), (c), (b), (g), (d), (e), (a)
3) (g), (a), (e), (c), (b), (d), (f)
4) (a), (c), (d), (g), (f), (b), (e)

Ans. 3

124. Match List-I with List-II.

	List-I		List-II
A)	Family	I)	Sapindales
B)	Genus	II)	Dicotyledonae
C)	Class	III)	Anacardiaceae
D)	Phylum	IV)	Angiospermae
E)	Order	V)	Mangifera

Choose the correct answer from the options given below:

- 1) A-II, B-III, C-V, D-I, E-IV
2) A-III, B-V, C-II, D-IV, E-I
3) A-I, B-V, C-II, D-IV, E-III
4) A-II, B-I, C-III, D-IV, E-V

Ans. 2

125. Arrange the following elements in descending order of their contribution to percentage weight of the human body.

- (a) Oxygen (b) Carbon
(c) Hydrogen (d) Nitrogen

Choose the correct answer from the options given below:

- 1) (b), (c), (d), (a) 2) (b), (a), (c), (d)
3) (a), (b), (c), (d) 4) (c), (a), (b), (d)

Ans. 3

126. Match List-I with List-II.

	List-I		List-II
A)	Cristae	I)	Flat membrane sacs in stroma of chloroplast
B)	Cisternae	II)	Infoldings in mitochondria
C)	Thylakoids	III)	Cell membrane
D)	Phospholipid	IV)	Disc shaped sacs in the Golgi apparatus

Choose the correct answer from the options given below:

- 1) A-II, B-IV, C-III, D-I 2) A-IV, B-III, C-I, D-II
3) A-III, B-IV, C-I, D-II 4) A-II, B-IV, C-I, D-III

Ans. 4

127. The number of action potentials generated by sino-arterial node (SAN) in a healthy human is _____ per minute.

- 1) 100-110 2) 120-140
3) 28-30 4) 70-75

Ans. 4

128. Symbiotic association between fungi and algae are called_____.

- 1) mycorrhiza 2) chrysophytes
3) lichens 4) sponges

Ans. 3

129. How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose?

- 1) 103 2) 412 3) 206 4) 309

Ans. 2

130. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans?

- 1) Femur-patella-tibia-tarsal
2) Femur-tarsal-patella-tibia
3) Femur-tibia-patella-tarsal
4) Patella-femur-tibia-tarsal

Ans. 1

131. Genus represents_____.

- 1) a group of closely related species
2) a group of closely related families
3) an individual plant or animal
4) a population of plants and animals

Ans. 1

132. The correct sequence of adult cell cycle phases is_____.

- 1) G1-S-G2-M 2) S-M-G2-G1
3) G1-G2-S-M 4) G1-M-G2-S

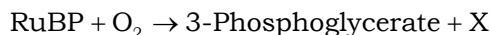
Ans. 1

133. Endomembrane system includes_____.

- 1) mitochondria, chloroplast, peroxisomes and vacuole
2) Golgi complex, chloroplast, peroxisomes and vacuole
3) endoplasmic reticulum, Golgi complex, lysosomes and vacuole
4) endoplasmic reticulum, chloroplast, peroxisomes and vacuole

Ans. 3

134. Photorespiration reaction catalyzed by RuBisCo is shown below:



Identify "X" from the given options:

- 1) Oxaloacetate
- 2) Malate
- 3) Phosphoenolpyruvate
- 4) 2-Phosphoglycolate

Ans. 4

135. Which of the following are characteristic features of Solanaceae family?

- (a) Flowers are bisexual and actinomorphic
- (b) Calyx have five sepals and are united
- (c) Androecium have five stamens and are epipetalous
- (d) Ovary is inferior

Choose the correct answer from the options given below:

- 1) (a) and (b) only
- 2) (b), (c) and (d) only
- 3) (a), (b) and (c) only
- 4) (d) only

Ans. 3

136. Sperm motility is due to_____.

- 1) amoeboid movement
- 2) muscular movement
- 3) flagellar movement
- 4) ciliary movement

Ans. 3

137. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R: Goats were more efficient at browsing than Abingdon tortoise.

In the light of the above statements, choose the most appropriate answer from the options given below:

- 1) A is correct but R is not correct
- 2) A is not correct but R is correct
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are correct but R is not the correct explanation of A

Ans. 3

138. Match List-I with List-II.

	List-I		List-II
A)	Both species are harmed	I)	Predation
B)	One species is harmed and the other is benefited	II)	Mutualism
C)	Both species are benefited	III)	Competition
D)	One is benefited while the other has no effect	IV)	Commensalism

Choose the correct answer from the options given below:

- 1) A-II, B-I, C-IV, D-III
- 2) A-III, B-I, C-II, D-IV
- 3) A-III, B-IV, C-II, D-I
- 4) A-I, B-II, C-III, D-IV

Ans. 2

139. Which of the following structure is not a part of the male reproductive system?

- 1) Vasa efferentia
- 2) Infundibulum
- 3) Rete testis
- 4) Epididymis

Ans. 2

140. Which of the following plant produces non-albuminous seeds?

- 1) Barley
- 2) Pea
- 3) Wheat
- 4) Maize

Ans. 2

141. Which of the following statements about lac operon is correct?

- 1) Genes i, z, y and a share single common promoter
- 2) Galactose can act as an inducer of lac operon
- 3) Gene i is constitutively expressed
- 4) Lactose activates repressor to bind to the operator

Ans. 3

142. Which of the following is used as an effective sedative and painkiller for treating post-surgery patients?

- 1) Morphine
- 2) Anti-retroviral drugs
- 3) Interferon
- 4) Antibiotics

Ans. 1

143. Which of the following is the correct order of arrangement of vertebrate column from the head to toe?

- 1) Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum
- 2) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum
- 3) Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebra
- 4) Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

Ans. 2

144. Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____.

- 1) cyclic guanine monophosphate
- 2) cyclic adenine monophosphate
- 3) acetyl choline
- 4) acetyl coenzyme A

Ans. 3

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Forelimbs of human and bats are homologous.

Reason R: Forelimbs of humans and bats have similar anatomical structure.

In the light of the above statements, choose the most appropriate answer from the options given below:

- 1) A is true but R is false
- 2) A is false but R is true
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are true, but R is not the correct explanation of A

Ans. 3

146. Given below are two statements:

Statement I: Down's syndrome is caused by the absence of one of the X-chromosomes.

Statement II: Turner's syndrome is caused by the presence of an additional copy of the chromosomes.

In the light of the above statements, choose the correct answer from the options given below:

- 1) Statement I is correct but Statement II is incorrect
- 2) Statement I is incorrect but Statement II is correct
- 3) Both Statement I and Statement II are correct
- 4) Both Statement I and Statement II are incorrect

Ans. 4

147. Which of the following statements about the reabsorption process in Henle's loop are correct?

- (a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.
 - (b) Urine gets concentrated in Henle's loop.
 - (c) Reabsorption of Na^+ and water takes place in Henle's loop.
 - (d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.
- Choose the correct answer from the options given below:

- 1) (a), (b) and (c) only
- 2) (a), (b) and (d) only
- 3) (a) and (b) only
- 4) (b), (c) and (d) only

Ans. 2

148. If the diploid chromosome number of typical angiosperm is 36, what would be the chromosome number in its endosperm?

- 1) 54
- 2) 72
- 3) 18
- 4) 36

Ans. 1

149. Arrange the following in descending order of number of species in the Amazonian rain forest.

- (a) Plants
- (b) Birds
- (c) Fishes
- (d) Invertebrates
- (e) Mammals

Choose the correct answer from the options given below:

- 1) (e) > (b) > (a) > (c) > (d)
- 2) (b) > (a) > (d) > (c) > (e)
- 3) (c) > (b) > (d) > (e) > (a)
- 4) (d) > (a) > (c) > (b) > (e)

Ans. 4

150. A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is _____.

- 1) 0.10
- 2) 0.99
- 3) 0.01
- 4) 0.02

Ans. 3

151. The opening between the right atrium and the right ventricle is guarded by

- 1) semilunar valve
- 2) sino-atrial node
- 3) bicuspid valve
- 4) tricuspid valve

Ans. 4

152. Which of the following enzymes synthesizes precursor mRNA?

- 1) RNA polymerase III
- 2) DNA polymerase
- 3) RNA polymerase I
- 4) RNA polymerase II

Ans. 4

153. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R: Isolation of genetic material needs disruption of cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- 1) A is correct but R is not correct
- 2) A is not correct but R is correct
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are correct but R is not the correct explanation of A

Ans. 3

154. Which of the following hormone is not secreted by human placenta ?

- 1) Progesterone
- 2) LH
- 3) hCG
- 4) Estrogen

Ans. 2

155. Colostrum, secreted by mother during initial days of lactation, is abundant in

- 1) IgA
- 2) IgD
- 3) IgG
- 4) IgM

Ans. 1

156. Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is

- 1) 10 million
- 2) 100 million
- 3) 1 million
- 4) 5 million

Ans. 1

157. Given below are two statements:

Statement I: Plasmids are autonomously replicating DNA.

Statement II: Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) Statement I is correct but Statement II is incorrect
- 2) Statement I is incorrect but Statement II is correct
- 3) Both Statement I and Statement II are correct
- 4) Both Statement I and Statement II are incorrect

Ans. 3

158. During PCR, primers bind to the DNA strands in the

- 1) annealing step
- 2) ligation step
- 3) denaturation step
- 4) extension step

Ans. 1

159. Match List-I with List-II.

List-I		List-II	
A	Transformation	I	Restriction enzyme
B	Cloning site	II	Transfer DNA to host bacteria
C	Selection	III	Replication
D	Ori	IV	Antibiotic

Choose the **correct** answer from the options given below:

- 1) A-III, B-IV, C-II, D-I
- 2) A-IV, B-I, C-III, D-II
- 3) A-II, B-I, C-IV, D-III
- 4) A-I, B-II, C-IV, D-III

Ans. 3

160. Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of

- 1) founder effect
- 2) genetic drift
- 3) divergent evolution
- 4) convergent evolution

Ans. 4

161. The covering of ovum at ovulation is

- 1) zona pellucida
- 2) chorion
- 3) endometrium
- 4) zona radiata

Ans. 1

162. Which of the following statements are **correct**?
a) Energy flow from producers to consumers is unidirectional

- b) Energy pyramid can never be inverted
- c) Transfer of energy follows the 1% law

Choose the correct answer from the options given below :

- 1) (a) and (c) only
- 2) (b) and (c) only
- 3) (a), (b) and (c)
- 4) (a) and (b) only

Ans. 4

163. How many theca are present in each lobe of a typical bilobed angiosperm anther ?

- 1) 8
- 2) 12
- 3) 2
- 4) 6

Ans. 3

164. Which of the following statements is **correct** about *Plasmodium*?

- 1) Gametocytes develop in mosquito gut
- 2) Fertilization takes place in mosquito gut
- 3) Reproduces sexually in liver cells
- 4) Reproduces sexually in RBCs

Ans. 2

165. For a person with blood group 'O' which of the following is **not** a possible combination of parents' blood group genotypes ?

- 1) Father: $I^A i$ and Mother: $I^B i$
- 2) Father: $I^A I^B$ and Mother: $I^A i$
- 3) Father: $I^A i$ and Mother: $I^B i$
- 4) Father: $I^A i$ and Mother: $I^A i$

Ans. 2

166. Which of the following is used as a clot buster?

- 1) Cyclosporin A 2) Statins
- 3) Streptokinase 4) Penicillin

Ans. 3

167. Which of the following disease is **not** sexually transmitted ?

- 1) Gonorrhoea 2) Genital warts
- 3) Syphilis 4) Tuberculosis

Ans. 4

168. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarf.

Reason R: Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) A is correct but R is not correct
- 2) A is not correct but R is correct
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are correct but R is not the correct explanation of A

Ans. 3

169. The method of directly injecting a sperm into ovum in assisted reproductive technology is called:

- 1) Intra cytoplasmic sperm injection (ICSI)
- 2) Embryo transfer (ET)
- 3) Gamete intra fallopian transfer (GIFT)
- 4) Zygote intra fallopian transfer (ZIFT)

Ans. 1

170. The inactive form of Bt toxin is converted to the active form in the insect gut

- 1) by proteases 2) by nucleases
- 3) due to alkaline pH 4) due to acidic pH

Ans. 3

171. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R: Resources are finite.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) A is correct but R is not correct
- 2) A is not correct but R is correct
- 3) Both A and R are correct and R is the correct explanation of A
- 4) Both A and R are correct but R is not the correct explanation of A

Ans. 3

172. Which of the following are secondary lymphoid organs ?

- a) Bone marrow b) Tonsils
- c) Spleen d) Thymus

Choose the **correct** answer from the options given below:

- 1) (b) and (d) only 2) (a) and (d) only
- 3) (a) and (b) only 4) (b) and (c) only

Ans. 4

173. Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs ?

- 1) Central cells 2) Polar nucleus
- 3) Antipodals 4) Synergids

Ans. 4

174. Given below are two statements :

Statement I: Ovulation is caused by LH surge leading to rupture of Graafian follicles.

Statement II: Graafian follicle remaining after ovulation transform into corpus luteum and secretes large amount of estrogen.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- 1) Statement I is correct but Statement II is incorrect
- 2) Statement I is incorrect but Statement II is correct
- 3) Both Statement I and Statement II are correct
- 4) Both Statement I and Statement II are incorrect

Ans. 1

175. Given below are two statements :

Statement I: Modern *Homo sapiens* arose in Australia and moved across continents.

Statement II: *Homo sapiens* arose around 75000 to 10000 years ago.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- 1) Statement I is correct but Statement II is incorrect
- 2) Statement I is incorrect but Statement II is correct
- 3) Both Statement I and Statement II are correct
- 4) Both Statement I and Statement II are incorrect

Ans. 2

176. Which of the following are primary consumers in a food chain ?

- 1) Herbivores 2) Carnivores
- 3) Parasites 4) Predators

Ans. 1

177. Which of the following is **not** evidence for evolution ?

- 1) Embryological support for evolution as proposed by Ernst Haeckel
- 2) Divergent evolution of anatomical structures such as forelimbs
- 3) Convergent evolution of traits like wings of birds and butterflies
- 4) Paleontological evidence from fossil records

Ans. 1

178. Sponges exchange O_2 with CO_2 by

- 1) tracheal tubes 2) gills
- 3) simple diffusion over their entire body surfaces
- 4) moist cuticle

Ans. 3

179. Match List-I with List-II.

List-I		List-II	
A	Excess growth hormone	I	Reabsorption of water and electrolytes in kidney
B	Luteinizing hormone	II	Contraction of uterus during child birth
C	Vasopressin	III	Acromegaly
D	Oxytocin	IV	Ovulation

Choose the **correct** answer from the options given below :

- 1) A-II, B-IV, C-I, D-III 2) A-IV, B-III, C-I, D-II
- 3) A-III, B-IV, C-II, D-I 4) A-III, B-IV, C-I, D-II

Ans. 4

180. Natural selection can lead to

- a) stabilisation b) genetic drift
- c) directional change d) disruption

Choose the **correct** answer from the options given below:

- 1) (a), (b), (c) and (d) 2) (a) and (c) only
- 3) (a) only 4) (a), (c) and (d) only

Ans. 4

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